GUIDE

erwin Data Modeler

Implementation Guide 15.0

Quest

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Contents

Overview 9

Database Design 9

About erwin Data Modeler 9

The Application Development Cycle 11

Customizing the User Interface 12

Customize Toolbars 12

Customize Display Levels 13

Setting Model-Level Default Values 14

Set Default Null Options, Data Types, Styles, and Object Owners 15

Set Default Referential Integrity Rules 17

Specify History Options for Model Objects and Events 20

Assign Model UDP Values 21

Using the Model Explorer 23

Overview of Model Explorer 24

Model View 26

Subject Area View 27

Create a Subject Area 29

Rename Objects 30

Select Unique Name Rules 31

The Go to Diagram Option 32

Add a Domain in the Model Explorer 33

Add a Model Source in the Model Explorer 34

Projects in erwin Data Modeler 34
Project Explorer Overview 35
The Theme Editor 38
Do you want to use an existing theme 41
Create and Apply a Theme 42
Create a Theme 43
Set Display Preferences for Entities and Tables 44
Set Display Preferences for Owned Attributes or Columns and Foreign Keys 46
Set Display Preferences for Keys 48
Set Display Preferences for Views and Materialized Views 49
Set Display Preferences for Subtype Symbol 51
Set Display Preferences for Annotations and Shapes 53
Apply a Theme to a Model 55
(Optional) Apply a Theme to Individual Objects in a Model 56
Apply a Theme by Binding a Model Template 57
Bind a Model Template 58
Creating and Using Domains 60
Overview of Domains 61
The Domain Editor 62
Drag a Domain from the Model Explorer to Your Model 64
Creating Domains 65
Add a Domain in the Logical Edit Mode 66
Add a Domain in the Physical Edit Mode 68
Assign a Data Type to a Domain 70
Assign or Change a Domain Icon 71

Applying Naming and Datatype Standards 72
Applying Naming Standards 73
Decide the Method 76
Create and Apply Naming Standards 77
Open NSM Option Editor from Model Explorer and Define Naming Standards for a Logical Model 78
Open NSM Option Editor from Model Explorer and Define Naming Standards for a Physical Model 79
Create Glossary Words and Abbreviations 80
Import Words and Abbreviations into the Naming Standards Glossary 82
Import Naming Standards from a Previous Version 83
Import from a Previous Version 84
Apply Naming Standards by Binding a Model Template 84
Apply Naming Standards by Binding a Model Template 85
Bind a Template to a Model 86
Verify Naming Standards Compliance 88
Apply Naming Standards to a Model 89
Example 90
Applying Naming Standards from Mart 93
Name Hardening Wizard 97
Override Name Inheritance 99
Unharden Object Names 100
Applying Data Type Standards 101
Decide a Method to Apply Standards 104
Create and Apply Data Type Standards 105
Define Data Type Standards from Model Explorer 106

(Optional) Create a Mapping 107
Apply Data Type Standards to the Model 108
Import Data Type Standards from a Previous Version 109
Import Data Type Standards 110
Apply Data Type Standards to the Model 111
Apply Data Type Standards by Binding a Model Template 112
Bind a Model Template 113
Create Reports Using Report Designer 114
Model Validation Reports 121
Pinned Reports from the Report Designer Menu 122
The Filter By Option 123
Report Designer Features 125
Subject Area Reports 128
Change the Logo on the HTML Report 129
Generating TLX Reports 130
Generate a Default TLX Report 130
Create a Custom TLX Report 130
Example--Number of Attributes in an Entity 136
Dimensional Modeling 138
Report on Data Sources, Target Columns, and Transformations 139
Report on Data Lineage with Diagrams 142
Supported Metadata Integration Bridges 145

erwin Product References

This document references the following products:

- erwin Data Modeler Standard Edition
- erwin Data Modeler Workgroup Edition
- erwin Data Modeler Navigator Edition

Overview

This section contains the following topics

Database Design
About erwin Data Modeler
The Application Development Cycle

Database Design

erwin Data Modeler (erwin DM) provides the tools to design and implement databases for transactional business, E-commerce, and data warehousing applications. You can create and maintain graphical models that represent databases, data warehouses, and enterprise data models. You can also define, manage, and implement corporate data requirements on several databases.

erwin DM includes a Windows-based graphical user interface, entity-relationship (ER) diagramming tools, custom editors, and a Model Explorer. The tools and editors help you define physical database objects and the Model Explorer provides a text-based view of model objects.

You can also use erwin DM to streamline the application development process. Data Architects, Business Analysts, Data Modelers, and so on, can work independently on different parts of a model or different model types. The groups can collaborate and synchronize their work as well.

About erwin Data Modeler

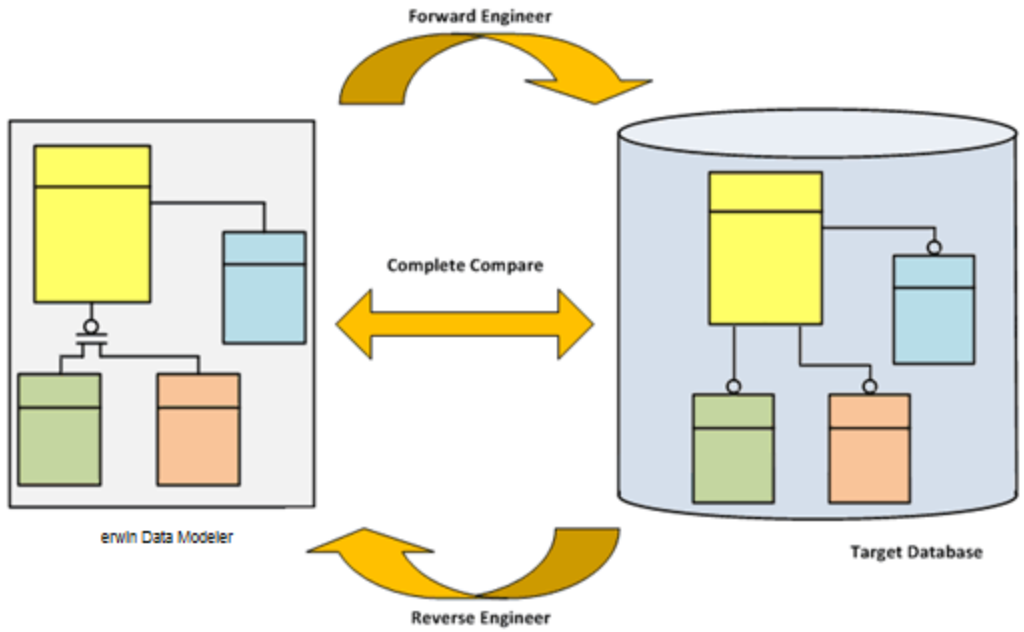
erwin DM lets you create and maintain your relational database and the logical and physical models that describe it. erwin DM works by providing a design solution that helps you create a visual blueprint (a data model) for your organization.

erwin DM helps you design a logical data model to capture business rules and requirements. The application also supports designing or deriving a corresponding physical data model for your target server. You can then forward engineer the physical data model and automatically generate the physical database structures to your system catalog.

erwin DM supports reverse engineering of existing databases and provides both a physical and a logical/physical data model. The forwarding engineering and reverse engineering options help you maintain an existing database, or migrate from your current database to a different one.

Overview

The Complete Compare feature automates the model and database synchronization by letting you compare the model with the database, displaying, and analyzing the differences. The comparison result enables you to move selected differences into the model or generate them into the database.



The Application Development Cycle

To create a data model, first the business rules are gathered and the logical constructs are created. Next, the physical design is created, and then a database is implemented. erwin DM supports this iterative process by supporting multiple platforms, reuse of objects, and synchronization of changes between data models.

Customizing the User Interface

This section contains the following topics

Customize Toolbars
Customize Display Levels

Customize Toolbars

Configure your work environment to display or hide a toolbar or display or hide the options on a toolbar. You can create a toolbar and can add only the options that you use frequently. You can also modify an existing toolbar to display or hide a subset of icons. For example, if you rarely use the underline and strikeout text formatting options on the Formatting Toolbar, you can hide those options.

Similarly, you can dock a toolbar in a specific position or keep it floating and move it as you want.

Follow these steps:

1. To modify an existing toolbar, follow these steps:
    a. Click the arrow-down icon on the icon toolbar that you want to customize.
    b. Select Add or Remove Buttons, <icon toolbar name> to display the currently selected icons.
    Your changes are reflected immediately in the erwin Data Modeler Workspace.
2. To create a toolbar, follow these steps:
    a. Click Tools, Customize.
    The Customize dialog opens.
    b. Go to the Toolbars tab and click New.
    c. Enter the name for your new toolbar.
    This text displays in the title bar for the new toolbar, when you select to undock it.
    d. Click OK.
    Your new toolbar name appears in the list of toolbars in the Customize dialog.
    e. Select the toolbar name, then click the Commands tab.
    f. Select a category, and drag the command icon out of the dialog to your new toolbar.
    Your new toolbar is populated with the commands that you choose.

Customize Display Levels

The Display Toolbar includes the options to change the display levels in your model. The following display levels are available:

- Entity or table
- Primary key
- Keys
- Attribute or column
- Comments

Display level is changed to suit the type of model being created or the audience for whom the model is being presented.

You can change the display level in two ways:

- Select an object or a group of objects, then click the desired icon to apply the display setting to those objects.
- Do not select any objects, then click the icon to apply the display setting to the entire diagram. If you have already set display options for individual objects, you can override those settings for the entire diagram with this option.

Setting Model-Level Default Values

This section contains the following topics

Set Default Null Options, Data Types, Styles, and Object Owners
Set Default Referential Integrity Rules
Specify History Options for Model Objects and Events
Assign Model UDP Values

Set Default Null Options, Data Types, Styles, and Object Owners

Use the Defaults tab in the Model Editor dialog to choose default values for null options, data types, themes, and object ownership.

Follow these steps:

1. Click Model, Model Properties.

   The Model Editor opens.

2. Click the Defaults tab and work with the following options:

   Non-Key Null Option

   Specifies whether you can have null as a value for a non-key attribute or column.

   Non-Identifying Relationship

   Specifies whether you can have null as a value for an attribute or a column that has a non-identifying relationship.

   Logical Datatype

   Specifies the data type that is used as the default for all attributes. You can change the data type for a specific attribute in the Attributes Editor dialog. You can also select whether the data type is inherited () or overridden ().

   > When you choose to inherit a data type, the value is inherited from the default settings. When you choose to override, the default value can be overridden and a different value can be added.

   Physical Datatype

   Specifies the default data type for all columns. You can change the data type for a specific column in the Columns Editor. You can also select whether the data type is inherent or overridden.

   Default Theme

   Specifies the default theme for the model. Your theme includes all the font, color, and formatting options for model objects. You use the Theme Editor to work with the formatting options.

Use Data Type in Unification

Specifies whether you want to use the data type when you unify three or more entities.

Migrated Column Options

Specifies the default settings for highlighting the primary and foreign keys on hovering over a relationship line. By default, the Display Diagram Highlights and Bold check boxes are selected. Also, the Color is set to the default one.

Enforce Relationship Rules

Specifies whether a supertype-subtype relationship between two different subtype entities belonging to same or different supertype group should be allowed. By default, the Enforce Relationship Rules check box is selected and such a relationship is not allowed.

Set Default Owners

Opens the Set Default Owner dialog where you can set a global default for all objects, or for individual objects. You enter an owner name, and can optionally reset the ownership for all objects when you click Set Owner For All Objects.

Important! Although you can quickly reset the ownership for all objects using this feature, you cannot undo this operation.

3. Click Close.

The default selections are set for the model and the Model Editor closes.

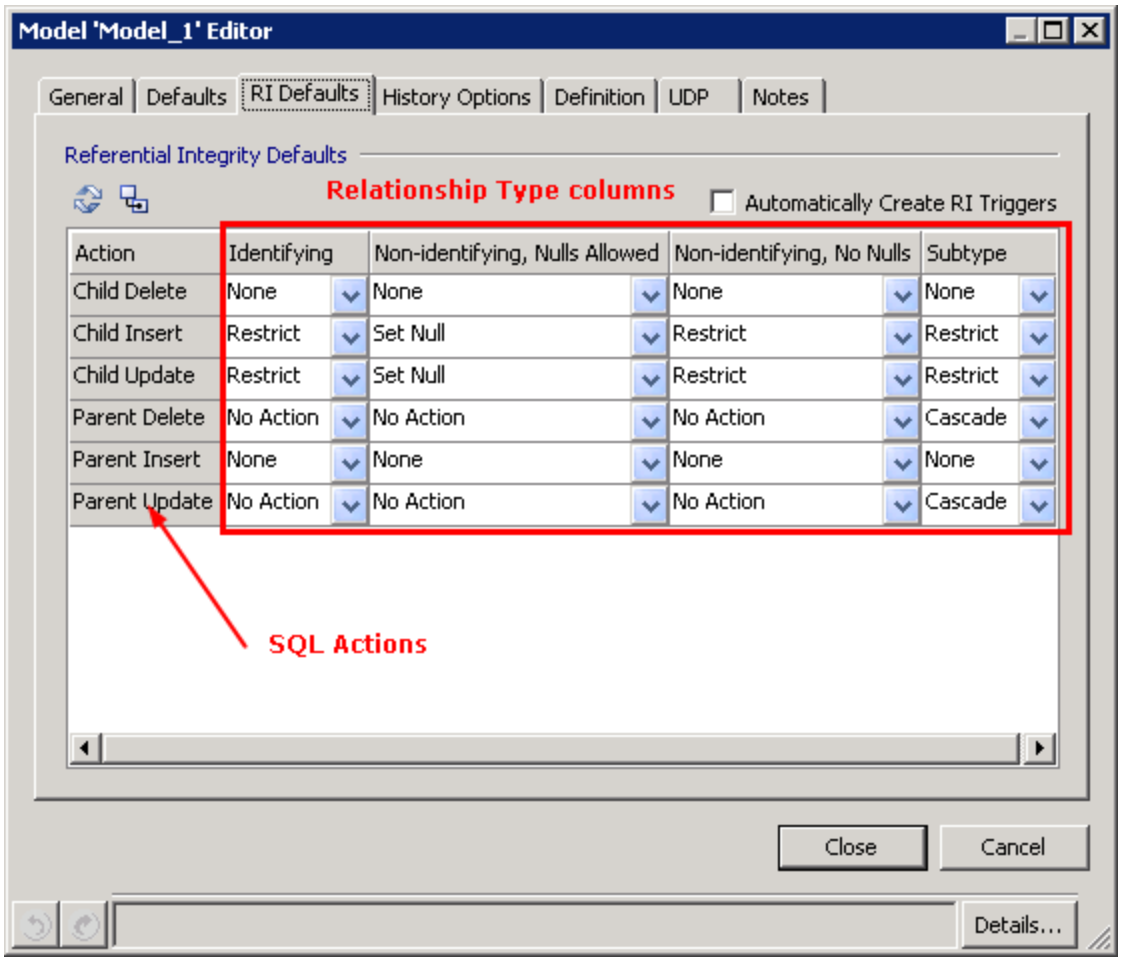
Set Default Referential Integrity Rules

Use the grid in the RI Defaults tab of the Model Editor to specify default referential integrity rules for relationships in your model.

Follow these steps:

1. Click Model, Model Properties.

   The Model Editor opens.



2. Click the RI Defaults tab and work with the following options:

Rebind referential integrity settings on all relationships

Applies the new default settings to any existing relationship in the model. If you change the default settings in an existing model, use this button to implement the new settings in the model for all existing relationships. If the new settings are saved, but not rebound, the defaults affect the new relationships only.

Reset all Referential Integrity default settings to their original values

Clears your settings and returns to the system defaults.

Automatically Create RI Triggers

Specifies whether you want to create triggers for the referential integrity defaults automatically.

Action

Lists a SQL action in each of its rows. For each SQL action, select the default RI action for each of the following relationship type columns:

Identifying

Lets you specify the RI setting for each of the SQL actions in the Action column. This setting applies to all identifying relationships in the model.

Non-identifying, Nulls Allowed

Lets you specify the RI setting for each of the SQL actions in the Action column. This setting applies to all non-identifying relationships in the model for which nulls are allowed.

Non-identifying, No Nulls

Lets you specify the RI setting for each SQL action in the Action column. The setting applies to all non-identifying relationships in the model for which nulls are not allowed.

Subtype

Lets you specify the RI setting for each of the SQL actions in the Action column. This setting applies to all subtype relationships in the model.

Some target databases, such as Oracle, use a different terminology for RI trigger actions. Use the RI setting appropriate for your target database. For NONE and NO ACTION, use

> of NONE means no restrictions (no RI trigger is created). NO ACTION is identical to RESTRICT. In other words, the actions are allowed only if the conditions are met.

3. Click Close.

   Default referential integrity rules are set and the Model Editor closes.

Specify History Options for Model Objects and Events

Use the History Options tab in the Model Editor to specify history preferences for common model objects and events. Maintaining history information is helpful if you want to review the history of actions that occur in your model.

Follow these steps:

1. Click Model, Model Properties.

   The Model Editor opens.

2. Click the History Options tab and work with the following options:

   History Objects

   Displays a list of the common model objects. Select the check box for each object for which you want to save history.

   History Events

   Displays a list of the model events. Select the check box for each event for which you want to save history.

3. Click Close.

   The history options are specified for the model and the Model Editor closes.

Assign Model UDP Values

After you create user-defined properties (UDPs) for a model, you can specify them in the UDP tab of the Model Editor.

Follow these steps:

1. Click Model, Model Properties.

   The Model Editor opens.

2. Click the UDP tab and work with the following options:

   Opens the User Defined Properties Editor so you can add a UDP or can edit an existing one.

   Resets the selected property to restore the inheritance.

   > This option is available only for property names that are inherited. In addition, this option becomes available only when you change the property value.

   Property

   Displays the name of the user-defined property. When you hover over a property name with your cursor, the UDP description displays as a tooltip.

   Value

   Specifies the current value that is assigned to the UDP. You can perform any of the following actions to change the value:

   - Enter the property value in the cell for *Integer*, *Real Number*, and *Text* data types.
   - Enter a date value for a *Date* data type.
   - Select a value from the Value list for the *List* data type. For example, you can add a UDP for a table to select a backup frequency such as *daily* from the Value list.

- Enter a system command or click ... to browse for a file or executable program for the *Command* data type. For example, you can enter C:\Myfile.doc or can select it to launch the Microsoft Word document *Myfile.doc*.

3. Click Close.

   Model UDP values are assigned and the Model Editor closes.

Using the Model Explorer

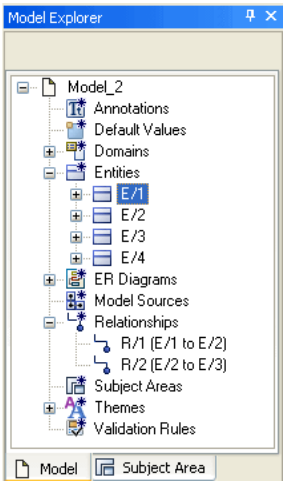
This section contains the following topics

Overview of Model Explorer
Model View
Subject Area View
Rename Objects
The Go to Diagram Option
Add a Domain in the Model Explorer
Add a Model Source in the Model Explorer

Overview of Model Explorer

The Model Explorer provides an organized, hierarchical, text-based view of your data model and its contents. It provides an easy method for creating objects. The Model Explorer lets you create, display, navigate, and modify your model.

The Model Explorer has two panes, Model and Subject Area, as shown in the following illustration:



All the objects in your model are listed in the Model Explorer, but display differently based on the active pane.

- The Model pane includes every object in your model including subject areas and domains. The Model pane is the default view.
- The Subject Areas pane displays model objects that are sorted by subject area.

When you change an object in the Model Explorer, the graphical view of the model is immediately updated with the same change. For example, if you rename a table in the Model Explorer, the new table name appears in the Diagram Window and the relevant editors. Similarly, if you change an object the diagram or in an editor, you immediately see the change in the Model Explorer.

Just as you can create an entity, a table, or a relationship using the Toolbox, you can create an object in the Model Explorer. To create an object, right-click on any object and click New. You can drag some objects from the Model Explorer onto the Diagram Window to create an object. For example, you can drag a domain onto an attribute or a column.

Using the Model Explorer, you can open the property editor for an object in two ways:

- Right-click the object and click Properties.
- Select the object and click the Property Editor icon in the Model Explorer toolbar.

> For bulk editing of object properties, you can use the Bulk Editor.

The Model Explorer and the data model are always in sync, regardless of the method you choose to add or edit an object.

You can also move, copy, and delete most objects in the Model Explorer, same as you do in a Diagram window.

After you move, copy, or delete an object, all the instances in the model that reference the object are automatically updated.

Model View

The primary purpose of the Model view is model exploration and to function as an alternative launching point for object and property editors. The Model view lists all the object types that appear in the current data model. The object types that are displayed here are based on the model type and the target server.

For example, if the model type is Logical, the Model Explorer does not include physical objects such as a view. Similarly, if the model type is Physical, the Model Explorer does not include logical objects. If the target server is Oracle, then objects specific to Oracle such as Clusters are displayed.

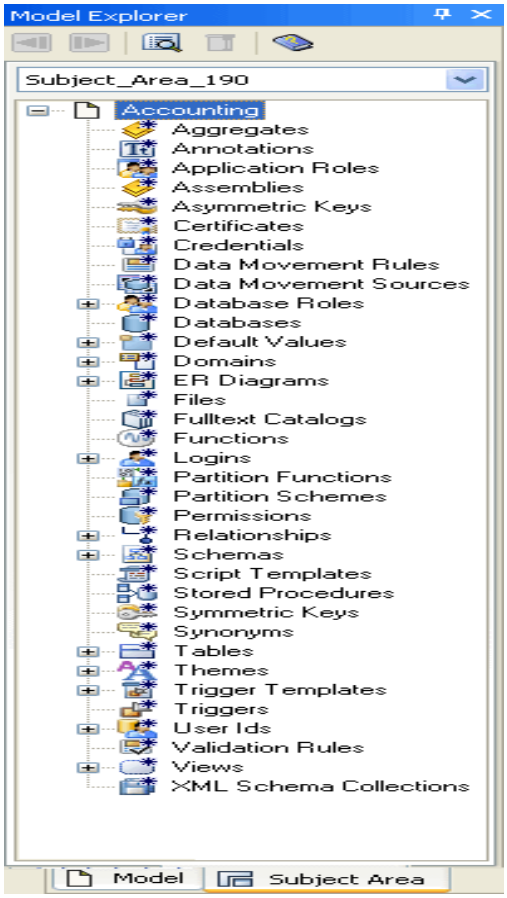
The following types of objects are not displayed in the Model Explorer:

- Minor objects that are included in the parent object editor. For example, a Valid Value object is created, deleted, and edited by the Validation Rule Editor.
- An object that represents configuration information, created for an internal process, and is not actual model data. For example, as a Complete Compare Option Set.

Subject Area View

The Subject Area view provides a filtered view of the model and minimizes unnecessary tree information. This view displays the model objects that are sorted by Subject Area. You can create Subject Areas to divide the model into smaller manageable parts, which is helpful when you work with large models.

The following illustration shows the Subject Area view:



Select a Subject Area to view the objects that are included in it. For example, in the previous illustration you see that the tables are categorized as tables, and views are categorized as views. This view is different from the display of the Subject Area objects in the Model view. Only those relationships are included here that are visible on the diagram and have both the endpoints present in the Subject Area.

The global objects that do not relate to the selected Subject Area, such as Subject Area objects or model source objects are not displayed.

Create a Subject Area

You can create a subject area using the Subject Area Editor.

Follow these steps:

1. Click Model, Subject Areas.The Subject Area Editor opens.
2. Click New.A new subject area with the default name is added to the Navigation Grid.
3. Work with the following options:

   Name
   Displays the subject area name.

   Auto-populate
   Specifies to populate the selected subject area with all the model objects automatically.
4. Click the General tab and specify common properties for the selected subject area.
5. Click the Members tab and specify the members for the selected subject area.
6. (Optional) Click the Definition tab to enter the definition text.
7. (Optional) Click the UDP tab to add subject area user-defined properties.
8. Click Close.
   A new subject area is created and the Subject Area Editor closes.

Rename Objects

Every object in your model is represented in the Model Explorer by its name. Just as you can rename an object using on-diagram editing in the Diagram Window, you can rename an object directly in the Model Explorer.

Follow these steps:

1. Click the object name, and then press F2.
   An edit box opens around the object name.
2. Enter the new object name.
   The new name of that object appears throughout the Model Explorer and the model.

> Any unique name rules that you select for your model are enforced when you rename an object in the Model Explorer.

Select Unique Name Rules

Unique name rules define how you want to respond to a duplicate entity or table name in the data model.

Follow these steps:

1. Click Tools, Model Naming Options.
   The Model Naming Options dialog opens.
2. Click the Duplicate Names tab and set preferences for duplicate names.
3. Click OK.
   Your changes are saved and the Model Naming Options dialog closes.

The Go to Diagram Option

When you have several hundreds of objects in a model, it might be difficult to locate a specific object. To locate an object in a diagram quickly, right-click the object in the Model Explorer and click Go To Diagram. The object is selected in the Diagram window.

Add a Domain in the Model Explorer

The Model view lists the domains for the current model, which include the default domains and the domains created by you.

Follow these steps:

1. Right-click a domain type (blob, string, number, and so on) and click New.
   An edit box opens with a default domain name.

   > The selected domain type is assigned as the parent.

2. Enter a name for the domain in the text box and press Enter.The new domain appears according to the sort order that you specify (either alphabetically or hierarchically). To switch the sort order of the Domains, right-click the Domains folder, and select the sort option that you prefer.

> For more information about domains, see Working with Data Models.

Add a Model Source in the Model Explorer

When you work with design layers, you link a model with its model source to identify the historical parent of the model. After you link a data model to its source, you can synchronize the two models to apply changes to one or the other selectively.

Follow these steps:

1. Right-click Model Sources and click New.
   The Add Model Source wizard opens.
2. Follow the steps in the Add Model Source wizard to finish adding the model source.
3. Click Add to complete the process and close the wizard.
   The source objects are added to your target model, and models are linked. As you continue to work with the models, all the changes you apply to the model and the model source are tracked.

For more information about adding model sources, see Working with Design Layers.

Projects in erwin Data Modeler

Projects in erwin Data Modeler helps you to bring all the database models involved in your application development together. Using projects, you can save and manage these models in a single location in your storage directory. The Project Explorer in erwin DM also enables you to open and load these models directly from a common explorer pane in the application, eliminating the need to load and work on them separately.

In projects, models are ordered in a hierarchical basis. You can create multiple child models from existing parent models in the project. You can choose whether these child models inherit the properties of their parent or start from a completely new ER diagram. Each model in the project can be worked on individually, with changes introduced exclusively to each other. However, erwin DM also lets you sync the changes in a parent model with all its child models, if needed. This helps to reduce the effort required to manually make the changes in all applicable models.

You can create new models as you require directly into the project. Projects also enable you to import and work on existing models that are stored in multiple locations. For example, using the Open Models function in the Project Explorer, you can import models from your local storage and erwin Mart Portal. Once they are imported, a copy of each model at the time of import is saved to

your specified project location. When you start working on the project, your changes are applied to these imported copies.

In addition to these methods of adding models, you can also use the Reverse Engineering function in projects to reverse engineer an existing script or database and add the resultant model to the project. Projects also enable you to generate forward engineering scripts for all the models in it. Furthermore, you can filter and customize the list of models to forward engineer, as you require.

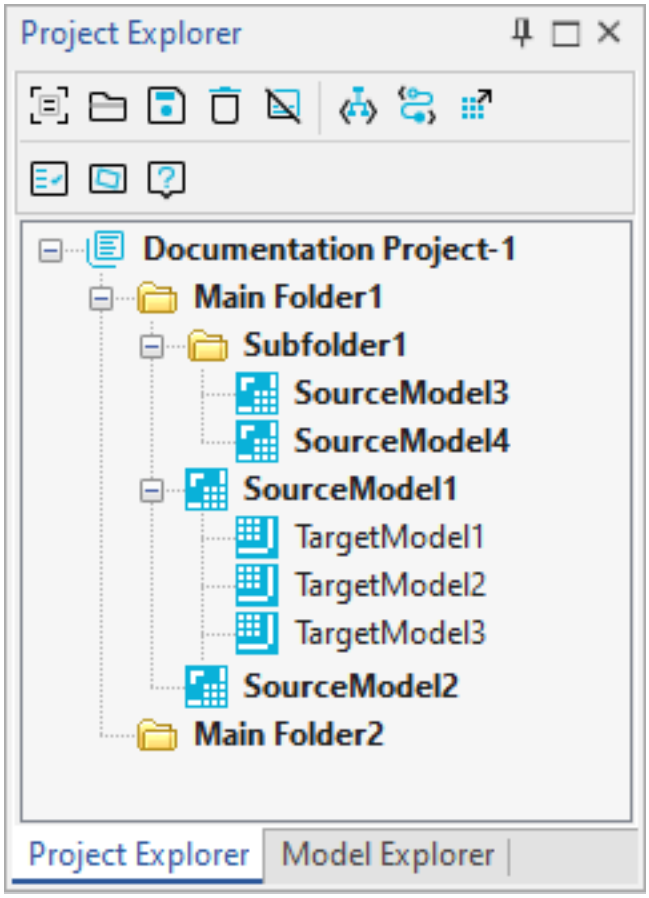
Projects can be saved as a single directory to any of your preferred location. You can save them to your local storage any time or upload the project files to a Mart catalog. When you save a project, all the models and their metadata are stored inside the specified project directory. You can launch the project from this storage directory using the project file, or individually open a model file saved in the project folders.

Project Explorer Overview

The Project Explorer provides an organized, ordered view of your project and the objects within it. You can use the Project Explorer to create new projects or edit existing projects by adding new models and modifying current models in the project.

The Project Explorer is divided into two sections: the Project Explorer toolbar and Project Explorer pane.

In the Project Explorer pane, database models can be organized within or outside folders and sub-folders. You can have as many folders and subfolders in a project, but a folder can have only one level of subfolders in it. The project and its contents are saved to erwin Mart Portal or your local machine in the same order and structure as specified in the Project Explorer pane.



The Project Explorer toolbar and the Project Explorer ribbon provides you with options to work on projects, such as:

New Project

Use this option to create a new project. For more information, refer to the Creating a Project topic.

Open Project

Use this option to open an existing project.

Save Project

Use this option to save the changes in the project.

Delete

Use this option to delete the selected objects in the project.

Close Project 

Use this option to close the project.

Schema Script 

Use this option to generate a forward engineering schema script for the objects in the project.

Alter Script 

Use this option to generate an alter script for the objects in the project.

Save to Mart 

Use this option to save the project folders and models to Mart.

Report 

Use this option to generate reports for the project. For more information, refer to the Project Reports topic.

Properties 

Use this option to edit the properties of the project or an object in it.

When you make changes to a model in the Project Explorer window, the changes are immediately updated in the diagram window. For example, if you rename a model in the Project Explorer, the new name replaces the existing model name in the diagram window.

The Project Explorer lets you perform the following actions:

- Create a new folder
- Create a new model
- Create a target model
- Rename a model
- Delete a model
- Sync child models with parent models
- Derive a model

For more information, refer to the Managing Projects topic.

The Theme Editor

A theme is a collection of font and color settings and is applied to a model, an object, a group of objects, or a subject area. Themes help you achieve the following objectives:

- Maintain the organizational design standard
- Denote the development lifecycle of a model
- Customize the designs in a model for various end users, for example, Human Resource or Marketing departments

Themes are applied in the following hierarchical order:

- Model - to replace the default theme that a new model uses.
- Subject Area - to override the theme of the model.
- Diagram - to override the theme of its owner (Model or Subject Area).
- Entity, View, Cached View, Annotation, or Attribute - to override the theme of the diagram.
- Drawing Object - to override all other themes. However, you cannot do an override on an attribute because attributes have no drawing objects. At the drawing object level, individual characteristics of a theme can be overridden once you select a theme.

You can open the Theme Editor in any of these ways:

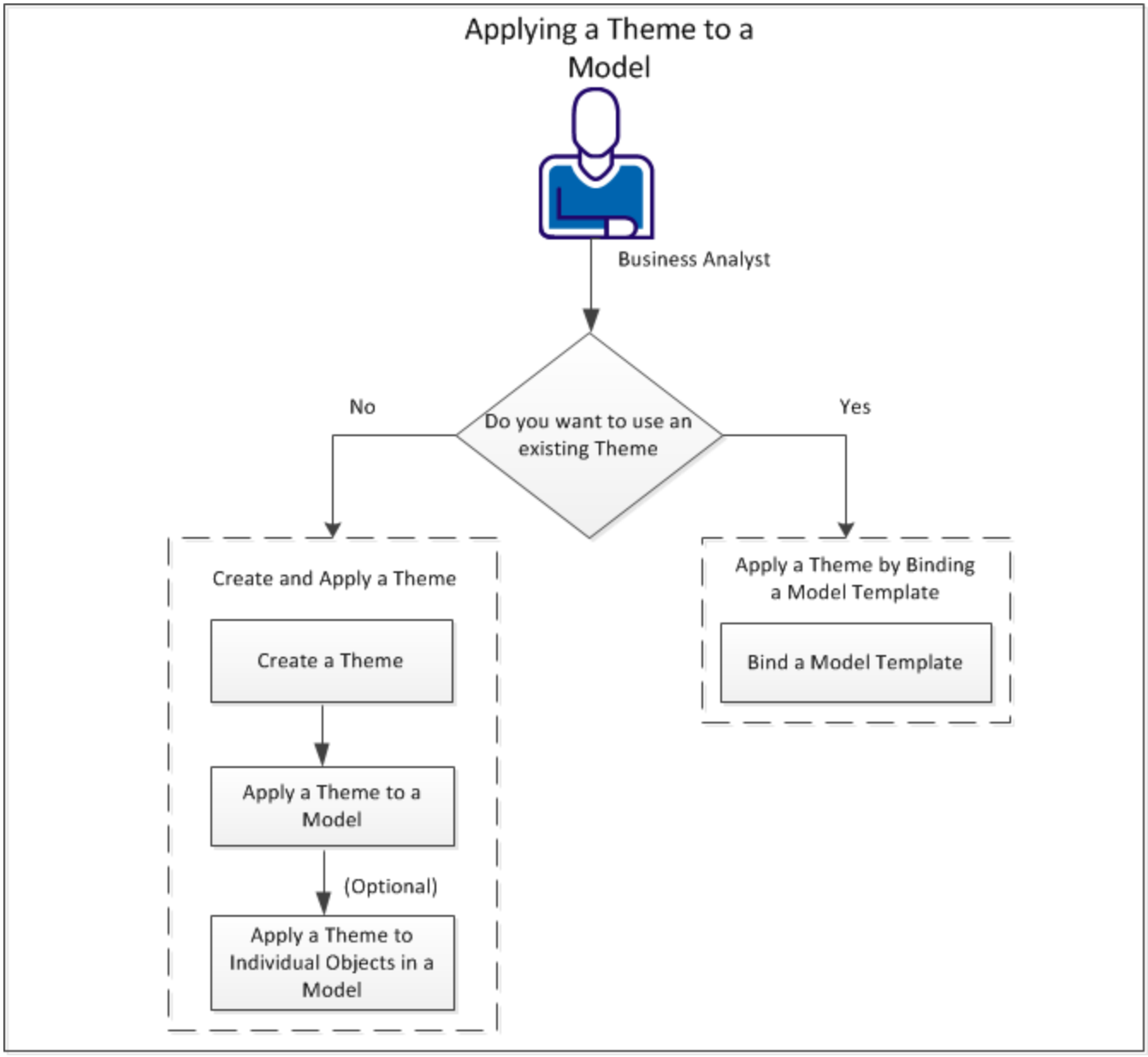
- From the General tab on the Diagram Editor
- When you click Themes on the View menu
- From the Model Explorer

The application ships with two themes, Default and Classic. You cannot edit these themes, but you can add new themes. You can also override the settings that are defined in a theme, in the Style tab of the individual editors.

For example, you are a Business Analyst in a software company. You want to do the following tasks:

- Maintain an organizational design standard in the Employee database.
- Organize the Employee database according to the departments in the company.

You identify all the departments in the company and create a theme for each department. You apply the themes to all of the departments to identify each department and its employees. This example is used consistently to explain all the procedures that are involved in creating and applying a theme.



Perform the following tasks to apply a theme:

- Do you want to use an existing theme
- Create and apply a theme
    1. Create a theme
    2. Apply a theme to a model

3. (Optional) Apply a theme to individual objects in a model

- Apply a theme by binding a model template

1. Bind a Model Template

Do you want to use an existing theme

If you have a standard theme in your organization, you can use that theme in all your models by binding it to a model template. Using a standard theme ensures consistency across all your models.

If you do not have any predefined themes, you can create a theme and can apply it to your models.

Back to Top

Create and Apply a Theme

If you do not have a standard theme in your organization, you can create one and can apply it for all your models. You can also create multiple themes and apply them to different objects in your model.

Create a Theme

After you open the Employee data model, identify the company departments and create a theme for each object or model.

When you create a theme, you can select and apply the theme to any object or model in the Theme Editor. For example, suppose that your Employee data model has separate objects for part-time and full-time employees working in the same department. You can apply a theme to the part-time employee object to differentiate it from the remaining objects in the Employee data model.

You use the Defaults tab to set the preferences for font and fill colors for all objects in your model. When you select a theme for your model, all objects in your model inherit these values.

Follow these steps:

1. Open erwin® Data Modeler and then open a data model.
2. Expand Model Explorer, right-click Themes, and click New.

   A new theme is created with a default name.
3. Right-click the new theme and click Properties.

   The Theme Editor opens.
4. (Optional) Edit the name of the theme.
5. Edit the properties in each column for the new theme.
6. Click the Defaults tab and make your selections for the desired options as appropriate for your organization.
7. (Optional) Customize your theme by selecting the desired options on the other tabs as appropriate for your organization.
8. Click Close.

   A theme is created and the Theme Editor closes.

Back to Top

Set Display Preferences for Entities and Tables

Use the Entity or the Table tab in the Theme Editor to set the preferences for displaying name and other definitions of entities or tables. The Entity tab is displayed when you view a Logical model and the Table tab is displayed when you view a Physical model.

Follow these steps:

1. Click View, Themes.

   The Theme Editor opens.

2. Select the theme for which you want to set the display preferences, and click the Entity or the Table tab.

3. Work with the following options:

   Name Display

   Font Options

   Specifies the default font styles to display the name of an entity. Click  to change the font style details. You can also choose to inherit, override, or harden this setting.

   Fill Options

   Specifies the default fill option for an entity. Click  to change the fill style details. You can also choose to inherit, override, or harden this setting.

   Outline Options

   Specifies the default outline styles for an entity. Click  to change the outline style details. You can also choose to inherit, override, or harden this setting.

   Definition Display

   Font Options

   Specifies the default font styles to display the entity definition. Click  to change the font style details. You can also choose to inherit, override, or harden this setting.

   Fill Options

   Specifies the default fill option for the entity definition. Click  to change the fill style details. You can also choose to inherit, override, or harden this setting.

   Outline Options

Specifies the default outline styles for the entity definition. Click  to change the outline style details. You can also choose to inherit, override, or harden this setting.

Your selections are displayed in the Sample section.

4. Click Close.

   Your changes are saved and the Theme Editor closes.

You cannot select to inherit, override, or harden any display setting for the Default theme.

Set Display Preferences for Owned Attributes or Columns and Foreign Keys

Use the Attribute or the Column tab in the Theme Editor to set the preferences for displaying attributes, columns, and foreign keys. The Attribute tab is displayed when you view a Logical model and the Column tab is displayed when you view a Physical model.

Follow these steps:

1. Click View, Themes.

   The Theme Editor opens.

2. Select the theme for which you want to set the display preferences, and click the Attribute tab.

3. Work with the following options:

   Owned Attribute Font

   Font Options

   Specifies the default font for entity attributes. Click  to change the font style details. You can also choose to inherit, override, or harden this setting.

   Foreign Key Inheritance

   Font Inheritance Type

   Specifies the type of inheritance for the font of a foreign key. You can choose the inheritance to be based on the primary key, the relationship, or no inheritance.

   Color Inheritance Type

   Specifies the type of inheritance for the color of a foreign key. You can choose the inheritance to be based on the primary key, the relationship, or no inheritance.

   Foreign Key Attribute Font

   Specifies the font of the foreign key attribute. Click  to select the font. You can also choose to inherit, override, or harden this setting.

   Foreign Key Attribute Color

   Specifies the color of the foreign key column. Click  to select the color. You can also choose to inherit, override, or harden this setting.

Your selections are displayed in the respective Sample sections.

4. Click Close.

   Your changes are saved and the Theme Editor closes.

You cannot select to inherit, override, or harden any display setting for the Default theme.

Set Display Preferences for Keys

Use the Key tab in the Theme Editor to set the preferences for displaying the alternate keys and inversion entities.

Follow these steps:

1. Click View, Themes.

   The Theme Editor opens.

2. Select the theme for which you want to set the display preferences, and click the Key tab.
3. Work with the following options:

   Key Display Preference

   Style Preference

   Specifies whether to show the alternate key (AK) style or the inversion entity (IE) style preference when both apply.

   Alternate Key (AK) Column Font

   Font Options

   Specifies the default font for the alternate key column. Click  to change the font style details. You can also choose to inherit, override, or harden this setting.

   Inversion Entity (IE) Column Font

   Font Options

   Specifies the default font for the inversion entity column. Click  to change the font style details. You can also choose to inherit, override, or harden this setting.

   Your selections are displayed in the respective Sample sections.

4. Click Close.

   Your changes are saved and the Theme Editor closes.

You cannot select to inherit, override, or harden any display setting for the Default theme.

Set Display Preferences for Views and Materialized Views

Use the View or the Materialized View tab in the Theme Editor to set the preferences for displaying the views and materialized views. The Materialized View tab is displayed only for the databases that support materialized views.

Follow these steps:

1. Click View, Themes.

   The Theme Editor opens.
2. Select the theme for which you want to set the display preferences, and click the View tab.
3. Work with the following options:

   Name Display

   Font Options

   Specifies the default font styles to display the name of a view. Click  to change the font style details. You can also choose to inherit, override, or harden this setting.

   Fill Options

   Specifies the default fill option for a view. Click  to change the fill style details. You can also choose to inherit, override, or harden this setting.

   Outline Options

   Specifies the default outline styles for a view name. Click  to change the outline style details. You can also choose to inherit, override, or harden this setting.

   Definition Display

   Font Options

   Specifies the default font styles to display the view definition. Click  to change the font style details. You can also choose to inherit, override, or harden this setting.

   Fill Options

   Specifies the default fill color for the view definition. Click  to change the fill style details. You can also choose to inherit, override, or harden this setting.

Outline Options

Specifies the default outline styles for the view definition. Click  to change the outline style details. You can also choose to inherit, override, or harden this setting.

Your selections are displayed in the Sample section.

Column/Attribute Display Font

Owned Column/Attribute

Specifies the default font style in an owned column or attribute. Click  to change the font style details. You can also choose to inherit, override, or harden this setting.

Expression

Specifies the default font style in an expression. Click  to change the font style details. You can also choose to inherit, override, or harden this setting.

The following fields are displayed only in the Materialized View tab:

Alternate Key (AK) Column/Attribute (for MV only)

Specifies the default font style in an alternate key column or attribute. Click to change the font style details. You can choose to make the font inherit, override, or harden.

Inversion Entity (IE) (for MV only)

Specifies the default font style in an inversion entry column or attribute. Click to change the font style details. You can choose to make the font inherit, override, or harden.

4. Click Close.

   Your changes are saved and the Theme Editor closes.

You cannot select to inherit, override, or harden any display setting for the Default theme.

Set Display Preferences for Subtype Symbol

Use the Subtype Symbol tab in the Theme Editor to set the preferences for displaying the subtype symbols.

Follow these steps:

1. Click View, Themes.

   The Theme Editor opens.

2. Select the theme for which you want to set the display preferences, and click the Subtype Symbol tab.

3. Work with the following options:

   Style Inheritance

   Inherit Discriminator font from Supertype name font

   Specifies that the font is inherited from the Supertype name font. If you select this check box, you cannot edit the Font Options field in the Subtype Symbol Text Font section.

   Inherit text color from line color

   Specifies that the text color is inherited from the line color. If you select this check box, you cannot edit the Color field in the Subtype Symbol Text Color section.

   Inherit Discriminator color from Supertype outline color

   Specifies that the line color is inherited from the parent Supertype outline color. If you select this check box, you cannot edit the Color field in the Subtype Symbol Style section.

   Subtype Symbol Text Font

   Font Options

   Specifies the font that you use for the subtype symbols in your model. If you have selected the Inherit Discriminator font from Supertype name font check box, you cannot edit this field.

   Subtype Symbol Text Color

   Color

Specifies the text color that you use for the subtype symbols in your model. If you have selected the Inherit text color from line color check box, you cannot edit this field.

Subtype Symbol Style

Line Width

Specifies the default line width for subtype symbols. Use the spin controls to increase or decrease the line width.

Color

Specifies the default color for subtype symbol lines. If you have selected the Inherit Discriminator color from Supertype outline color check box, you cannot edit this field.

Fill Options

Specifies the default line fill options for subtype symbols. Click  to change the fill style details.

> You can select to inherit, override, or harden any of these settings except for display settings for the Default Theme.

4. Click Close.

   Your changes are saved and the Theme Editor closes.

Set Display Preferences for Annotations and Shapes

Use the Annotation/Shape tab in the Theme Editor to set the font and color of the annotation text. Use this tab to change the display properties of annotations and shapes.

Follow these steps:

1. Click View, Themes.

   The Theme Editor opens.

2. Select the theme for which you want to set the display preferences, and click the Annotation/Shape tab.

   Font Options

   Specifies the default font styles for the selected drawing object. Click  to change the font style details. You can also choose to inherit, override, or harden this setting.

   Vertical Alignment

   Specifies the vertical alignment position of the text in the annotation. You can also choose to inherit, override, or harden this setting.

   Horizontal Alignment

   Specifies the horizontal alignment position of the text in the annotation. You can also choose to inherit, override, or harden this setting.

   Text Handling Style

   Specifies how you want the annotation to adjust itself when you enter text. You can also choose to inherit, override, or harden this setting.

   Fill Options

   Specifies the default fill color for the selected annotation. Click  to change the fill style details. You can also choose to inherit, override, or harden this setting.

   Line Options

   Specifies the default line styles for the selected annotation. Click  to change the outline style details. You can also choose to inherit, override, or harden this setting.

   Outline Options

   Specifies the default outline styles for the selected annotation. Click  to change the outline style details. You can also choose to inherit, override, or harden this setting.

Click OK.

The new style for the annotation is saved.

Apply a Theme to a Model

Apply the theme to the Employee data model and maintain the organizational design standard. When you apply a theme to a model, all new objects inherit the customized values that you have defined in the theme.

Follow these steps:

1. Click Model, Model Properties.

   The Model Editor opens.

2. Click the Defaults tab.
3. Click the Default Theme drop-down list and select a theme.

   Your theme is applied to the model.

4. Click Close.

   The Model Editor closes.

You have successfully applied a theme to your model.

If you want to apply a different theme to the individual objects in your model, see the next topic. You do not have to close the model to apply a different theme.

Back to Top

(Optional) Apply a Theme to Individual Objects in a Model

You can apply a theme to individual objects instead of the entire model. In your Employee data model, you can apply a different theme to each department (object) to organize the model. For example, you can apply one theme to the Human Resources object and another theme to the Marketing Division object.

You can also apply a theme by copying it from one object and pasting it to another object using the context menu. You can copy and paste a theme between similar objects and not between different objects. For example, you can copy and paste a theme between entities and not between an entity and a diagram. This option is useful if you want to apply a theme including any overrides that you have applied for a specific entity.

Follow these steps:

1. Ctrl + click each object that you want to select in your model.
2. Right-click and click Override Fonts and Colors.

   The Multiple Shape Object Editor opens.
3. Click the Style tab.
4. Click the Theme drop-down list under the Theme for Default Font and Color options section, and select your theme.

   Your theme is applied to the selected objects.
5. Click Close.

   The Multiple Shape Object Editor closes.

The theme is applied to the selected objects.

Back to Top

Apply a Theme by Binding a Model Template

If you have a standard theme in your company, you can bind a model template having the standard theme to any new data model. When you bind the model template, the current model inherits the theme from the model template.

Bind a Model Template

Binding a model template to a data model copies the non-built-in objects from the template model to the target model. When you bind a template to a model, the predefined objects populate your model. This process eliminates the need to create objects manually or to use reverse engineering. Changes to the model template are displayed in the models that use it, maintaining consistency of your object names. When you open a model that is bound to a model template, you can choose to synchronize any model template changes with current the model. Whenever you open a model that is bound to a model template, confirm whether to synchronize any model template changes with the objects used in the model.

You can bind model templates any time, while creating a model or after the model is created.

Objects that are brought in to a model from a template fully follow the modeling rules. If any conflicts occur in the model, the existing data is preserved as much as possible.

Follow these steps:

1. Open erwin Data Modeler and then open a data model.
2. Click File, Model Templates, Bind Template.

   The Bind Model Template dialog opens.
3. Use the toolbar to browse for the template file locally on your computer or in the Mart.
4. Select one of the following options and then click OK:

   Load the entire contents of the template

   Specifies to load the entire contents of the template. This option binds all of the model template objects to the current model without the opportunity to review and select individual template objects.

   Run the Template Wizard to select objects for synchronization

   Specifies to open the Model Template Synchronization Wizard so that you can select the objects in the model template to synchronize with the data model.

   The model opens in the workspace with the template objects copied into it. The theme is applied to the data model.

While binding a template using this method, synchronization on loading the model does not work as expected for inherited templates. As a workaround, do either of the following:

- Refresh the template using the **File** > **Model Templates** > **Refresh Template** option.
- Select **Preserve the template binding** while creating a new model.

Back to Top

Creating and Using Domains

This section contains the following topics

Overview of Domains
The Domain Editor lists all the domains for the data model.
Drag a Domain from the Model Explorer to Your Model
Creating Domains
Assign a Data Type to a Domain
Assign or Change a Domain Icon

Overview of Domains

A domain is a model object that you can use to assign properties quickly to an attribute or column. By using domains, you promote consistency because you can reuse a domain as many times as you like in a single or multiple data models.

Domains also reduce the time that is spent on development and maintenance. If you change the domain, all attributes or columns that are associated with the domain also change.

Domains have two types of properties:

- **Non-inheritable properties**: The non-inheritable properties do not migrate to child domains or attributes and columns that are associated with the domain. This is because they are properties of the domain itself.
- **Inheritable properties**: The inheritable properties migrate to child domains and to the attributes and columns that are associated with the domain.

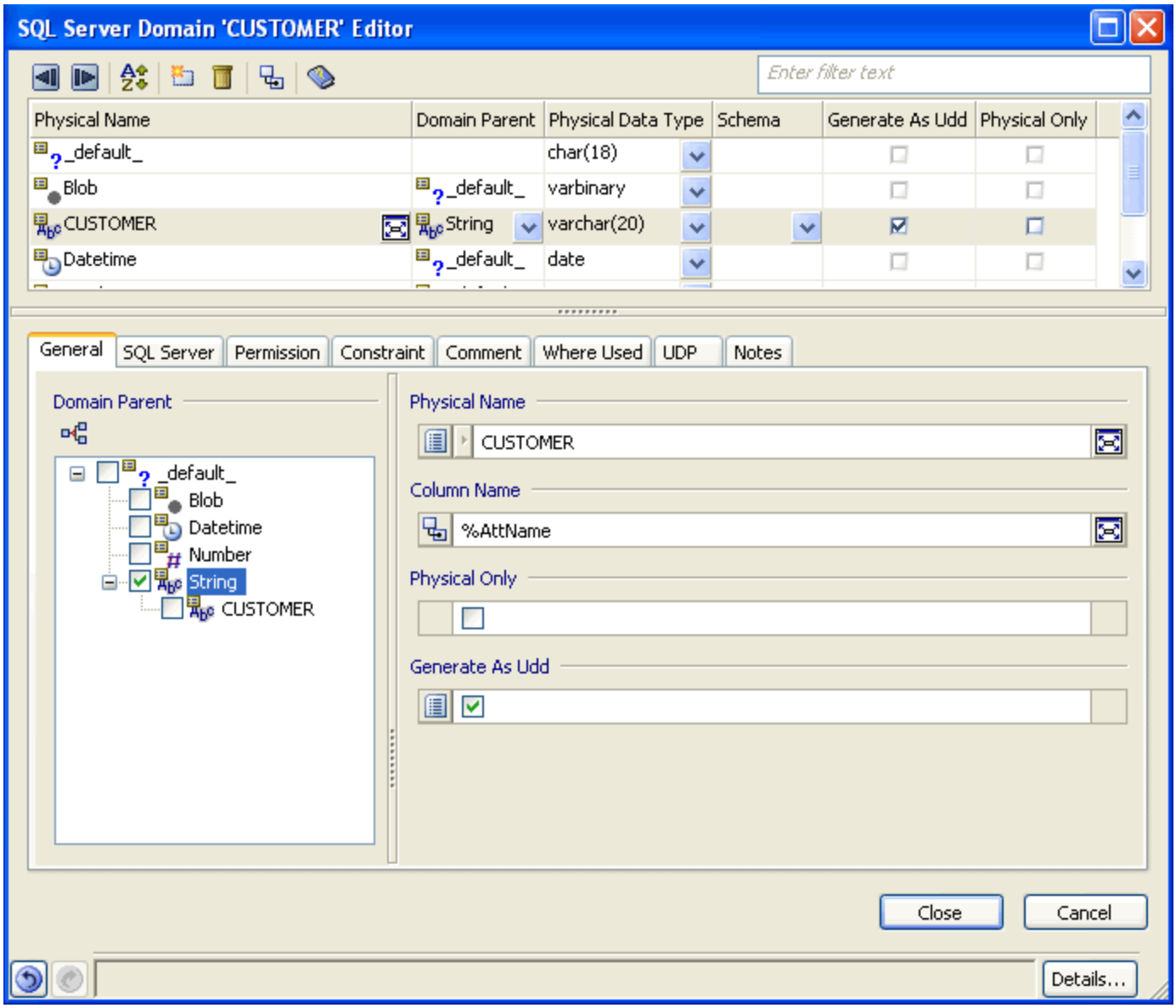
The inheritable and non-inheritable domain name properties appear on the General tab. They also appear on the target database tab (for example, SQL Server) in a physical model.

When you drag-and-drop a domain from the Model Explorer onto an entity or a table, you add an attribute or a column. The new attribute or column inherits all the inheritable properties, including name, from the domain.

In the Model Explorer and in the Domain Editor, a unique icon represents each domain. A default icon is used for new domains, but you can choose a different icon to represent the domains you create.

The Domain Editor

The Domain Editor lists all the domains for the data model. You can arrange them hierarchically or alphabetically. The Model Explorer and the Domain Editor list the same domains and lets you sort the domains in hierarchical or alphabetic order. You can create and modify both physical and logical domains using the Domain Editor.



The tabs and options in the editor change depending on whether the model is logical, physical, or logical/physical.

Domains include the following properties:

- Domain name and column name
- Column data type, permission, default value, and valid value
- Domain comment or note and column comment or name
- User-defined property
- User notes

> You cannot change the properties of the standard domains (<default>, Blob, Datetime, Number, and String).

Drag a Domain from the Model Explorer to Your Model

Use drag-and-drop to add a domain from the Model Explorer to an entity or a table in your model.

Follow these steps:

1. Expand the list of domains in the Model Explorer and click the domain that you want to add to your model.
2. Drag and drop the domain onto the entity or table in your model.
   The domain is added as an attribute or column.

   > Depending on the Unique Naming rules that apply to the data model, you are prompted to rename each attribute or column. You are prompted because, the same domain name is assigned to each new attribute or column automatically.

Creating Domains

Use the Domain Editor to create domains in addition to the default domains. You can create a domain while you are in a logical model (logical edit mode) or a physical model (physical edit mode). The options in the editor change depending on the edit mode.

Add a Domain in the Logical Edit Mode

The Domain Editor lets you define properties for attributes and columns. You can create a domain and can attach it to any number of attributes and columns instead of defining properties for each column individually. You can work with features in the Domain Editor to add a domain to the logical model.

Follow these steps:

1. Click Model, Domains.

   The Domain Editor opens.

2. Click the New button on the toolbar.

   A new domain with a default name is added and is placed in alphabetical order.

3. Work with the following options in the Navigation Grid:

   Name

   Displays the name of the domain in the logical model.

   > Duplicate domain names are not allowed.

   Domain Parent

   Displays the current parent domain. You can change the domain using the drop-down list.

   Logical Data Type

   Specifies the logical data type for the domain. You can change the data type using the drop-down list. The drop-down list contains the data types related to the current domain.

   Logical Only

   Specifies whether the domain appears only in the logical model. Select this check box if you want the domain to appear only in the logical model. Clear this check box if you want the domain in the physical model. If you select this check box in a logical-only model and you derive a new physical-only model, the domain does not appear in the new model.

4. Work with the other tabs in the editor to define more domain properties.

5. Click Close.

   The domain is added to the logical model and the Domain Editor closes.

Back to Top

Add a Domain in the Physical Edit Mode

You can work with the features in the Domain Editor to add a domain to the physical model.

Follow these steps:

1. Click Model, Domains.

   The Domain Editor opens.

2. Click the New button on the toolbar.

   A new domain with a default name is added and is placed in alphabetical order.

3. Work with the following options in the Navigation Grid:

   Physical Name

   Displays the name of the domain in the physical model.

   Domain Parent

   Displays the current parent domain. You can change the domain using the drop-down list.

   Physical Data Type

   Specifies the physical data type for the domain. You can change the data type using the drop-down list. The drop-down list contains the data types related to the current domain.

   Anchored to Type

   Specifies whether the domain is anchored to a column or a variable. This column is enabled only for a Db2 LUW user-defined data type.

   Schema

   Specifies the schema that is associated with the domain.

   Distinct Type

   Specifies whether the domain is of distinct type.

   Physical Only

   Specifies whether the domain appears only in the physical model. Select this check box if you want the selected domain to appear in the physical model only. Clear this check box if you want the selected domain in the logical model.

4. Work with the other tabs in the editor to define more domain properties.

5. Click Close.

   The domain is added to the physical model and the Domain Editor closes.

Back to Top

Assign a Data Type to a Domain

A data type is a domain property and defines the physical properties of a column in a database. You can define length (number of characters), type (alpha or numeric), and precision (decimal) properties. You can also define a logical data type, which usually corresponds to a physical data type.

By default, a new domain is assigned the same data type as its parent domain. You can change it at any time.

Follow these steps:

1. Click Model, Domains.
   The Domain Editor opens.
2. Select the domain in the Navigation Grid for which you want to assign a data type.
3. Select a new data type for the domain using the drop-down list in the Logical Data Type column (logical edit mode) or the Physical Data Type column (physical edit mode).
4. Click Close.
   The data type is assigned to the domain and the Domain Editor closes.

Assign or Change a Domain Icon

You can assign any image (.bmp file) as the icon for the non-inheritable property of the domain. You can assign the same image or can select a different image that the attributes or columns of the domain can inherit.

> You can change the domain icon only in the logical model.

Follow these steps:

1. Click Model, Domains.
   The Domain Editor opens.
2. Select the domain in the Navigation Grid that you want to work with and click the General tab.
3. Specify the image to use in the Image drop-down or click New to open the Image Editor and create one.
4. Click Close.
   The icon is assigned to the selected domain.

Applying Naming and Datatype Standards

This section contains the following topics

Applying Naming Standards
Applying Data Type Standards

Applying Naming Standards

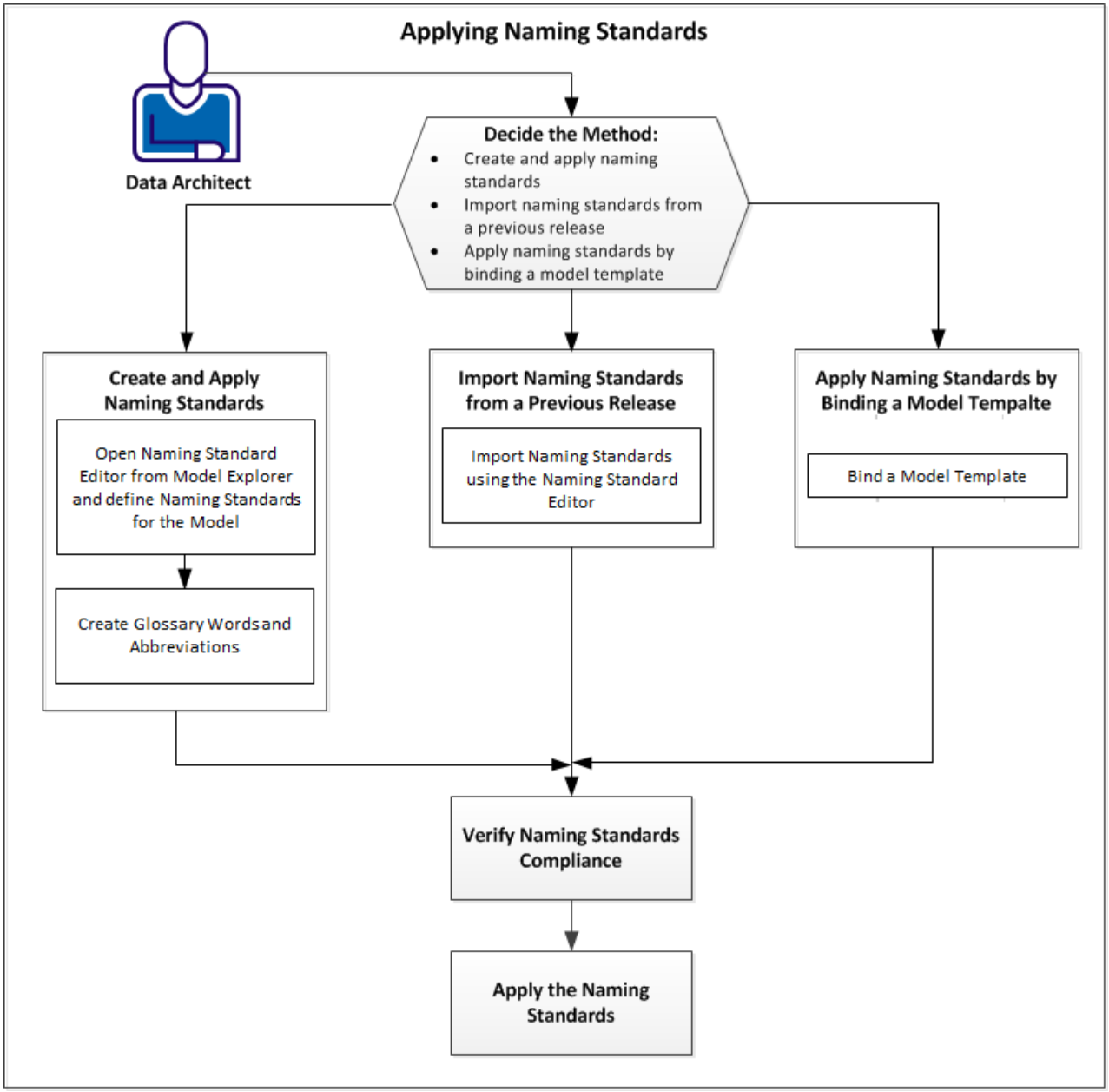
You create naming standards so that everyone in your organization uses names correctly and consistently. Use the Naming Standard Editor to create naming standards and develop a glossary that contains glossary words, their abbreviations, and alternate abbreviations.

You can create multiple Naming Standard objects and can apply one of them to a model. Each Naming Standard object includes the naming standards for logical and physical model objects, and the glossary. The components of logical and physical model names are prime, modifier, or class. To define naming standards for entities, tables, attributes, columns, and domains specify the components and the order in which to present them.

If you do not use naming standards, you may have modeling design problems. These design problems could produce errors when you run queries to extract data.

> Suppose that you upgrade a model from release r7 through r7.3.12 or from r8 through r8.2.9 to the latest release. If the source model includes NSM files, the naming standards are not included in the upgraded model. To get the naming standards, save the NSM file to your local computer, import it to a template model, and then bind the template to the upgraded model. We recommend that you do not include any other objects in the template model, other than the NSM object.

You can apply naming standards in three ways, as illustrated in the following diagram:



Apart from the three ways of applying naming standards explained in the above diagram, you can also apply naming standards from the Mart.

Follow these steps apply naming standards:

1. Decide the method
2. Verify naming standards compliance

3. Apply the naming standards

Decide the Method

You can apply naming standards in the following ways:

- Create and apply: Use this option when you are creating the naming standards for the first time.
- Import from a previous version: Use this option if you have created naming standards using a previous version of erwin DM.
- Bind a model template: Use this option if you want to use the naming standards that you have already created and attached to a model or a model template.
- Connect to the Mart and use NSM files available in NSM File Management.

Create and Apply Naming Standards

You create naming standards if you are implementing standards for the first time.

Open NSM Option Editor from Model Explorer and Define Naming Standards for a Logical Model

You can specify naming standards for logical names that are different from the naming standards for physical names. Use the Logical tab in NSM Option Editor to apply naming standards to entities, attributes, and domains. In addition, use this tab to specify the order in which parts of names appear in the logical name.

Follow these steps:

1. Expand the Model Explorer, right-click Naming Standard and click New.

   A new Naming Standard object is added.

2. Right-click the new object and click Properties.

   The Naming Standard Editor opens.

3. To apply the naming standard to your model:
   - In the list of naming standards, select your naming standard and then select Is Active.

4. To preserve any changes made to the naming standard in the Physical mode and reflect those in the Logical mode:
   - In the list of naming standards, select your naming standard and then select Physical to Logical.

   > - The physical mode to logical mode naming standard implementation caters to cases where the logical definition is absent.
   > - For Hive, when reverse engineering (REC or RES) is done, the Physical to Logical option on the Naming Standard Editor is not set by default. You need to set it manually.

5. Go to the Logical tab.
6. Select the part of the name such as prime, modifier1, modifier2, or class that you want to include first in the logical name. Specify the parts in the Entity, Attribute, and Domain rows.
7. Click Close.

   The data is saved.

Back to Top

Open NSM Option Editor from Model Explorer and Define Naming Standards for a Physical Model

You can specify naming standards for physical names that are different from the naming standards for logical names. Use the Physical tab in Naming Standard Editor to apply naming standards to tables, columns, and domains. In addition use the editor and specify the order in which parts of names appear in the physical name.

Follow these steps:

1. Expand the Model Explorer, right-click Naming Standard and click New.

   A new Naming Standard object is added.

2. Right-click the new object and click Properties.

   The Naming Standard Editor opens.

3. To apply the naming standard to your model:
   - In the list of naming standards, select your naming standard and then select Is Active.

4. To preserve any changes made to the naming standard in the Physical mode and reflect those in the Logical mode:
   - In the list of naming standards, select your naming standard and then select Physical to Logical.

     - The physical mode to logical mode naming standard implementation caters to cases where the logical definition is absent.
     - For Hive, when reverse engineering (REC or RES) is done, the Physical to Logical option on the Naming Standard Editor is not set by default. You need to set it manually.

5. Go to the Physical tab.
6. Select the part of the name such as prime, modifier1, modifier2, or class that you want to include first in the physical name. Specify the parts in the the Table, Column, and Domain rows.
7. Click Close.

   The data is saved.

Back to Top

Create Glossary Words and Abbreviations

You can enter words and their corresponding abbreviations directly into the glossary. You can enter, display, edit, and specify the word type (prime, modifier1, modifier2, or class).

Follow these steps:

1. Expand the Model Explorer, right-click Naming Standard and click New.

   A new Naming Standard object is added.

2. Right-click the new object and click Properties.

   The Naming Standard Editor appears.

3. Complete the following options in the NSM Option Editor:

   Name

   Specifies the default NSM name. You can modify the NSM name when required.

   Match whole words only

   Specifies whether to substitute glossary abbreviation on whole words only or to include parts of words or substrings. Select the check box to substitute glossary abbreviation on whole words only. Clear the check box to substitute glossary abbreviation for complete words, as well as on parts of words or substrings.

   Match whole name only

   Specifies whether to substitute glossary abbreviation on whole name only. Select the checkbox to substitute glossary abbreviation on whole name only and to ignore parts of words and substrings.

   For example, a glossary abbreviation, Customer Name is substituted with its abbreviation only when the whole name is found. The parts of words or substrings, Customer and Name are ignored.

   Is Active

   Specifies whether to apply a NSM to a model.

   Physical to Logical

   Specifies whether the changes made to the Naming Standard in the Physical mode should be preserved and reflected in the Logical mode.

- For Hive, when reverse engineering (REC or RES) is done, the Physical to Logical option on the Naming Standard Editor is not set by default. You need to set it manually.
- As a best practice, do not select the Physical to Logical check box while deriving a PO model; that has NSM objects or NSM template attached to it; to LO model. Earlier an option for Physical to Logical NSM was not available. However, as it is in erwin Data Modeler now, you need not use NSM for deriving a new model. For example, PO to LO.

4. Click the Glossary tab and work with the following options:

   Word Type

   Displays only the selected type of glossary words.

   Word/Words

   Specifies the glossary word or words that you want to abbreviate.

   Abbreviation

   Specifies the standard abbreviation.

   Alt Abb

   Specifies an alternate abbreviation.

   P, M1, M2, C

   Specifies whether you want to use the word as a prime, modifier 1, modifier 2, or class.

5. Click  to save the glossary as a .CSV file. You can use this file at a later point, when you perform reverse engineering and want to preserve the naming standards attached to the model.
6. Click Close.

   The data is saved.

In case of duplicates, the glossary word/abbreviation that you created the latest will be applicable.

Back to Top

Import Words and Abbreviations into the Naming Standards Glossary

You can import words and abbreviations into the glossary from an existing naming standards CSV file. Be sure to examine the data in each row and edit or add as required.

Follow these steps::

1. Expand the Model Explorer, right-click Naming Standard and click New.

   A new Naming Standard object is added.

2. Right-click the new object and click Properties.

   The Naming Standard Editor appears.

3. Click the Glossary tab, then click Import.

   The Open dialog appears.

4. Select the path and file name of the CSV file, and click Open.

   The words and corresponding abbreviations are imported into the Naming Standards glossary.

Back to Top

You have two tables, Customer and Accounts. The Customer table has First Name and Last Name columns.

Suppose you want to implement the following standards:

- Prefix the names of the tables with Tbl_.
- Display First Name as Col_FNAME and Last Name as Col_LNAME.

How you do it:

1. Add a Naming Standard Object and make it active.
2. In the Logical tab, for Entity and Attribute, select Modifier 1 in the Part 1 and Prime in the Part 2 column.
3. In the Physical tab, for the Table and Column object types, select Modifier 1 in the Part 1 and Prime in the Part 2 column.
4. In the Glossary tab, add First Name and Last Name and the corresponding abbreviations. Select the P check box.
5. Open the Model Naming Options dialog and click the Name Mapping tab.
6. Select the Use Glossary check box for Entity to Table and Attribute to Column rows. In the

Prefix column, enter Tbl_ in the first row and Col_ in the third row. Click Close.

The standards that you have defined are applied to the model.

Import Naming Standards from a Previous Version

If you have created naming standards in a previous version of erwin DM, you can import them to the current release. When you import naming standards, the definitions for logical and physical models, and glossary words are imported.

Import from a Previous Version

If you have created naming standards for a previous version of erwin DM, you can import them to an NSM Option object. You import a .NSM file.

Follow these steps:

1. Expand the Model Explorer, right-click Naming Standard and click New.

   A new Naming Standard object is added.

2. Right-click the new object (or an existing object) and click Properties.

   The Naming Standard Editor appears.

3. Click Import Other.

4. Select the .NSM file and then click Open.

   The NSM file is displayed as a new Naming Standard object. The standards for Logical and Physical models are displayed in the respective tabs. The glossary words and corresponding abbreviations are displayed in the Glossary tab.

Back to Top

Apply Naming Standards by Binding a Model Template

When you bind a model template to the current model, the naming standards are also applied. If you run the wizard to select individual objects to bind, select the Naming_Options, NSM_Option, and NSM_Glossary_Word check boxes.

Apply Naming Standards by Binding a Model Template

When you bind a model template to the current model, the naming standards are also applied. If you run the wizard to select individual objects to bind, select the Naming_Options, NSM_Option, and NSM_Glossary_Word check boxes.

Bind a Template to a Model

Binding a model template to a data model copies the non-built-in objects from the template model to the target model. When you bind a model template, the predefined model objects appear in your model. Binding eliminates the need to enter objects manually or depend on reverse engineering. Changes to the model template are reflected in the models that are bound to it, maintaining consistency of object names. Whenever you open a model with a bound model template, you can synchronize any model template changes with the objects used in the model.

Binding of model templates can occur at any time necessary, including during model creation, or after the model is created.

Objects that are brought in to a model from a template fully respect modeling rules. If any conflicts occur in the model, the existing data is preserved as much as possible.

Follow these steps:

1. Click File, Model Templates, Bind Template.

   The Bind Model Template dialog opens.

2. Select the model template to bind to the current model using the drop-down control. Use the toolbar to browse for the template file locally on your computer or one that is stored in a mart.

3. Select one of the following options to specify how to bind the template, and click OK.

   Load the entire contents of the template

   Specifies to load the entire contents of the template. This option binds all of the model template objects to the current model without the opportunity to review and select individual template objects. The NSM Option object that is active in the model template is made active in the current model as well.

   Run the Template Wizard to select objects for synchronization

   Specifies to open the Model Template Synchronization Wizard so that you can select the objects in the model template to synchronize with the data model.

   The model opens in the workspace with the template objects copied into it.

While binding a template using this method, synchronization on loading the model does not work as expected for inherited templates. As a workaround, do either of the following:

- Refresh the template using the **File** > **Model Templates** > **Refresh Template** option.
- Select **Preserve the template binding** while creating a new model.

Back to Top

Verify Naming Standards Compliance

To know if the naming standards that you have created are applied properly, use the Naming Standards Compliance dialog.

Follow these steps:

1. Click Tools, Standards, Check Naming Standards Compliance.

   The Check Name Standards Compliance dialog opens.

2. Select the objects to check, select the Check Word Order and Position check boxes, and then click Start.

   All instances of non-compliance are displayed in the Check Name Standards Compliance dialog. You can replace the name, replace all instances of the name, or ignore the non-conforming name.

Back to Top

Apply Naming Standards to a Model

After you define naming standards using the NSM Option Editor, select individual model objects and apply the standards. The glossary words and abbreviations (or alternate abbreviations) that are defined in the active NSM Option object are applied as naming standards to a model.

Follow these steps:

1. Open a model that includes an active NSM Option object.
2. Click Tools, Model Naming Options.

   The Model Naming Options dialog opens.
3. Click the Name Mapping tab.
4. For each model object type that you want to use the glossary, select the Use Glossary check box and select the type of abbreviation. Click OK.

   The naming standards are applied when you switch from a logical model to a physical model.

   In a Logical/Physical model, define the naming standards, create the abbreviations, and select the object that uses the glossary in the Logical view. Switch from the Logical view to the Physical view and the standards are applied.

   In a Logical Only (LO) or a Physical Only (PO) model, define the naming standards, create the abbreviations, select the object that uses the glossary, and then derive a PO or an L/P model. In the Derive Model wizard, go to Naming Standards. Select the model that has an active NSM object as the active model template. The naming standards are applied to the derived model.

> The translation from logical to physical through the NSM Option Object works only if the physical name is inherited from the logical name. That is, only if the names have not been manually changed in the physical model. If a physical name is modified in a logical/physical model, the inheritance from the logical side to the physical side is overridden. The naming standards no longer work. However, you can reset the override property to inherit from the logical name to restore this inheritance.

NSM File Attached to an Older Version Model

When you open a model from an older version of the product, it is upgraded to the current release. If an NSM file was attached to the model, the file is not imported and attached to the model automatically. Import the NSM file to a template and attach the template to the model manually.

Back to Top

Example

Use this example to create database specific templates and standardize database specific properties across all your data models. If you want to include the Naming Standards in each database specific target template, you must change the naming standards glossary list for each template. You must change the glossary for each template, because the Naming Standards information is stored within a model. This section describes how you can manage Naming Standards from a single source.

Follow this process:

- Create a logical-only model template with general model set up criteria and include the Naming Standards information.
- [Optional] Create a database specific template and use this template as the binding source and select the ♦Synchronize on Model Load♦ option.
- Attach this template as the binding template source for your model and select the Synchronize on Model Load option.

  If you have to change something in the glossary, you just change the Naming Standards object in the logical model template and save the template. When you open a model, if the auto-synchronize option is enabled, the database specific template model is opened. Next, the logical model template is opened and the Naming Standards object is synchronized with the database specific template. The remaining template information is also populated in the new model.

Example

This section includes two examples with two different databases♦Db2 and Oracle. The examples use templates that are stored in a Mart.

1. Create a logical model template that includes the following basic standards desired for all data models:
   - A standard theme.
   - An Annotation that enables the model name to be presented on diagrams (a type of legend).
   - Standard Domains.
   - Standard Entity level User Defined Properties (UDP) that can be shared among all data models.

- A set of start-up diagrams. Each diagram includes a diagram shape to enable the presentation of the diagram name in each diagram.
- A Naming Standards object.

2. Open this model and save it to a Mart as a template; provide the name ﷺTmplGeneralﷺ.

   To create each database specific template in erwin:

   a. Click File, New.

   b. Select Type as Logical/Physical.

   c. Select the Target Database as Oracle.

   d. Select TmplGeneral for Model template.

   e. Select Preserve the template binding.

   You now have a logical/physical model with all the pre-populated information from the TmplGeneral logical model template. You can add any common Db2 characteristics that you want all your Db2 models to share.

In addition, make the following changes:

1. Click Model, Model Properties and change the model name to TmplDb2.
2. Click Model Explorer, Annotations - Change the Annotation to reflect the model name:
3. Perform the following steps to prevent the over writing of model names during synchronization:

   a. Click File, Model Template, Run the Template Wizard. The Model Template Synchronization Wizard appears.

   b. Go to the Model page and change the action of the Annotation object named ﷺModelLegendﷺ to Ignore. This setting bypasses synchronizing the model name.

   c. In the Settings page, select the Ignore synchronization of the root model object option. This setting prevents over writing of the model level properties from the logical only model.

   d. Click Finish.

4. Click Tools, Model Naming Options. Verify whether the Use Glossary option in the Name Mapping tab is set for the model objects for which you want to leverage the naming standards glossary.
5. Also, as these diagrams were synchronized from a logical only model, go to the Diagram Properties dialog for the Physical Overview and Relational Key diagrams and change the View Mode to ﷺPhysicalﷺ.

6. Save this model to your mart library as model name ♦TmplDB2♦ (with the Mark as Template option)
7. Repeat these steps for each database specific template (i.e. SQL Server, Oracle, Teradata, etc.). With this, you can add any database specific properties in your templates to use across all application data models.

   You can now use these database specific templates for your application data models. Either create a new model, or, bind an existing data model to the database specific template model manually. If you are binding an existing model, review the model to verify that no properties in the template model will cause issues with the existing model when synchronized.

In this example, creating a new application data model using the template is used as a starting point.

1. Select File, New.
2. Select Match template.
3. Select the TmplDB2 template model from the Mart.
4. Select Preserve the template binding option.
5. Click OK.

   Now you have a Db2 application data model with all the properties from both the TmplGeneral template and the TmplDB2 template.
6. If you use the Annotation ModelLegend text box, select Ignore in the template. Selecting ignore prevents this object from being updated during synchronization.
7. Save this model to your Mart library. To follow this example, save it as model ♦AppModelDB2♦.
8. To test the synchronization process, add a few Entities in the ♦Logical Overview♦ diagram and drag/drop the domain ♦Identification♦ as primary keys to both Entities. (Select the Identification domain and drag it to the primary key locations in each Entity).
9. Select the ♦Physical Overview♦ diagram. Notice that the word Identification changed to its abbreviation from the Naming Standards Glossary.
10. Save and Close the model as AppModelDB2.
11. Open the original model template TmplGeneral and change the abbreviation for Identification from ♦Id♦ to ♦Idnt♦.
12. Save and close this model.

13. Open model AppModelDB2.

    Notice that the abbreviation is automatically applied through the template hierarchy to this application data model for Db2.

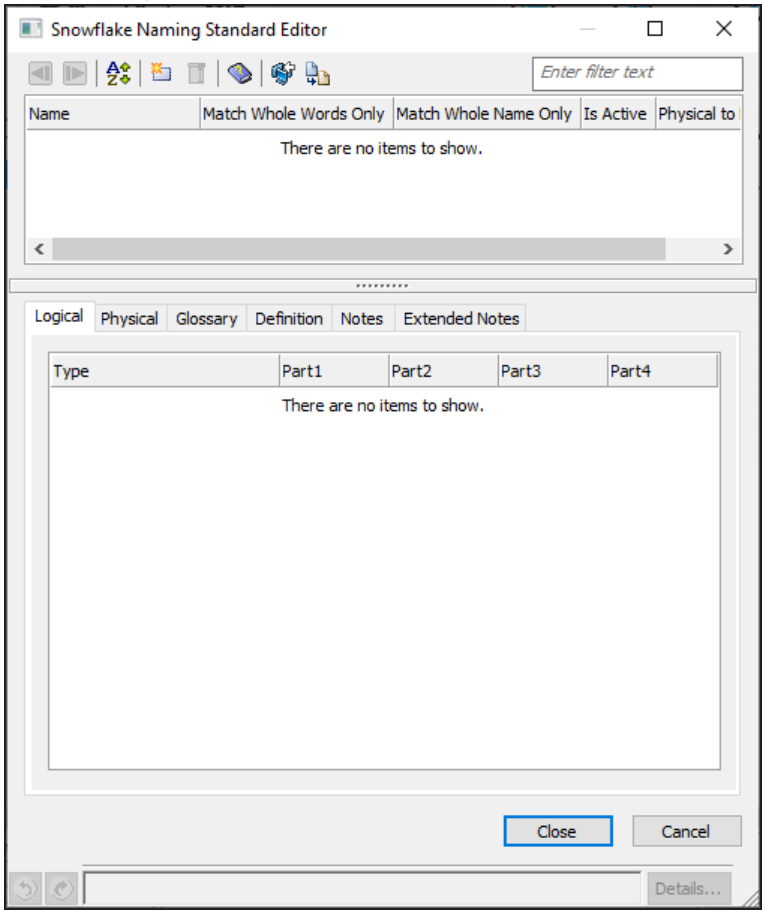
Applying Naming Standards from Mart

You can store NSM files centrally on erwin Mart and apply them to both, local and Mart models from the Naming Standard Editor.

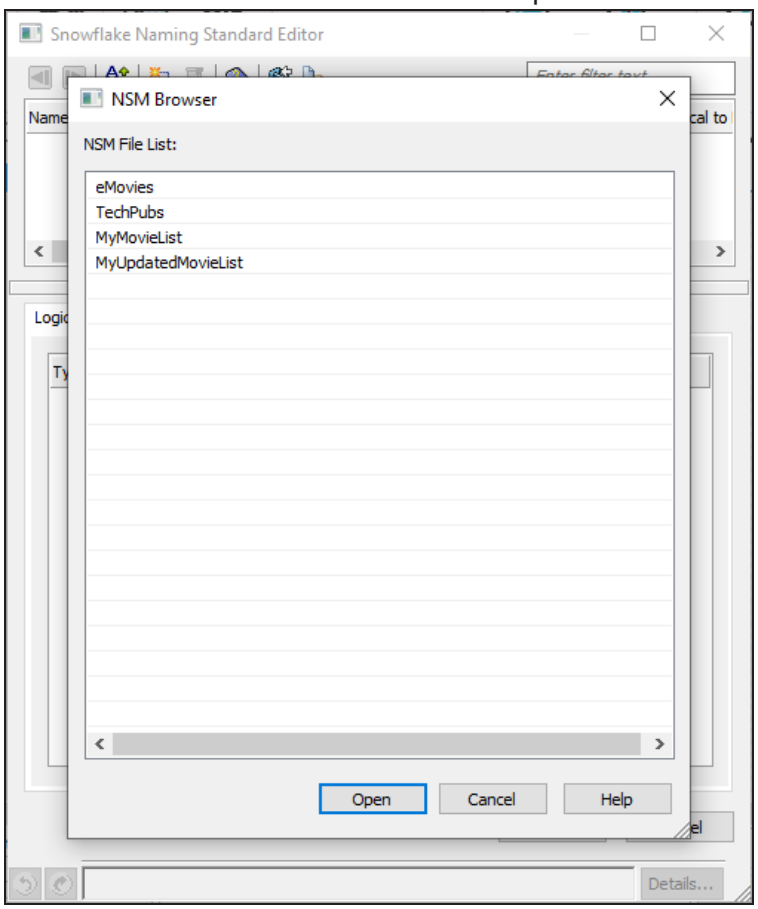
To apply naming standards from the Mart, follow these steps:

1. Connect to the mart.
2. Open a model (local or Mart model).

3. Click **Tools** > **Naming Standards**.
   The Naming Standard Editor opens.



4. On the toolbar, click  .
   The list of NSM files available on the mart opens.



5. Select an NSM file and click **Open**.
   The selected NSM file is applied to your model.

You can sync any changes made to the applied NSM file from the Mart to erwin Data Modeler (DM).

To sync the applied NSM file, follow these steps:

1. On the Naming Standard Editor, click the **Glossary** tab.
2. On the toolbar, click  .

> If you import the NSM file to erwin DM and apply it to a model via the Naming Standard Editor; any updates made to the NSM file in the erwin Mart Portal do not sync on the erwin Data Modeler side. NSM sync works when an NSM file is applied to a model via erwin Mart Portal.

Name Hardening Wizard

The Name Hardening Wizard lets you override inheritance of names and block name changes or resets within a model. Using the wizard, you can select the object types and object instances for which you must harden logical or physical names, or both. The wizard helps you override the name inheritance for all instances of the selected object types, a subset of those instances, or a specific instance.

> erwin Data Modeler is designed to convert Logical Names to Physical Names using macros. When you convert Physical Name to a Logical Name in a model, Name Hardening cannot be performed.

To run the Name Hardening Wizard, go to **Tools** > **Manage Name Hardening**.

You can harden names for the following object types:

- Attribute/Column
- Default value
- Domain
- Key group/Index
- Relationship
- Entity/Table
- Validation rule

> If you prefer, you can select all the objects directly on the diagram, launch the wizard, and go directly to the Action page. The objects you select on the diagram populate the Type Selection and Object Selection pages.

The Name Hardening Wizard provides three options: Type Selection, Object Selection, Action. Using the the three options you can perform the following action:

- Override Name Inheritance
- Unharden Object Names

More information:

Override Name Inheritance
Unharden Object Names

Override Name Inheritance

You can override the inheritance of names and can harden the names. If a name is hardened, the physical model name does not change upon changing the logical name.

Follow these steps:

1. Go to **Tools** > **Manage Name Hardening**.

   The Name Hardening Wizard opens.

2. Click **Type Selection**.

   The options that you select on this page determine the options that are defined on the Object Selection page.

   > While you can select the Entity check box in a logical model, you can harden names only in a physical model. Logical entity names are not inherited names; they are set on the entity object. Therefore, they do not change and there is no need to harden them.

3. Select the object types for which you want to override inheritance, then click **Object Selection**.
4. Select the model objects for which you want to override inheritance, then click **Action**.

   If some of the subordinate objects are selected, then the check box for that collection node is selected and is also shaded.

5. Select the check box for each item for which you want to override inheritance.
6. Click **Apply**, and then **Close**.

   The Logical and Physical columns indicate the status of the objects and the Selected Objects column lists their fully qualified names. For example, instead of simply 'a,' an attribute name would be 'OwnerName.E/1.a.'. When you unharden an object name, the same name you used in the hardening process remains. However, now you can edit or reset that name (editing or resetting of hardened names is not allowed).

   > Unhardening an object name does not automatically restore inheritance, it only allows the names to be reset or changed.

More information:

Name Hardening Wizard

Unharden Object Names

The Name Hardening wizard can be used to unharden names. When you unharden an object name, the same name you used in the hardening process remains. After the unharden process is complete, you can edit or reset the name (editing or resetting of hardened names is not allowed).

> Unhardening an object name will not automatically restore inheritance, it only allows the names to be reset or changed.

Follow these steps:

1. Go to **Tools** > **Manage Name Hardening**.

   The Name Hardening Wizard opens.
2. Click **Type Selection**.
3. Select the object types that you want to unharden, then click **Object Selection**.
4. Select the model objects that you want to unharden, then click **Action**.
5. Clear the check boxes for objects that you want to unharden.
6. Click **Apply**, and then **Close**.

   The name for each object that you selected is unhardened, and the Name Hardening Wizard closes.

More information:

Name Hardening Wizard

Applying Data Type Standards

A data type is a predefined set of characteristics for an attribute or column that specifies field length, acceptable characters, and optional and required parameters. For example, the data type Char(18) specifies that the column can store up to 18 alpha and numeric characters.

By default, a data type is applied to every attribute in a logical model and to every column in a physical model. In the logical model, you assign the data type or the domain from which the attribute inherits its properties determines it. In the physical model, you assign the data type or the default value that is specified by the target server determines it.

Considering the large quantity of attributes or columns a data model usually has, it is tedious to assign and maintain consistency of data types manually. The DSM Option Editor helps you easily assign and maintain data types in a model.

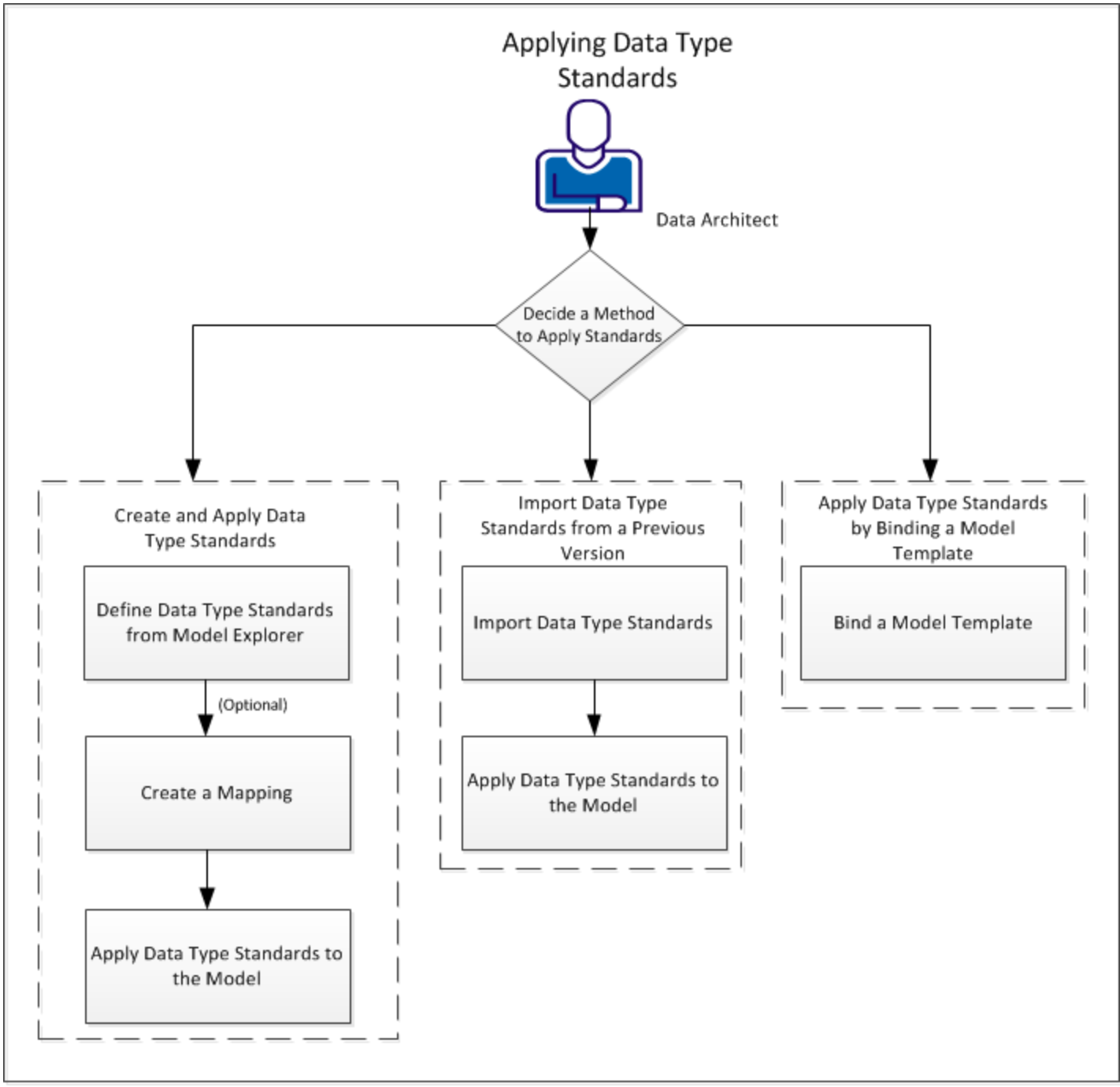
Create data type standards so that everyone in your organization uses data types consistently.

When you create a DSM Option object, it is preloaded with default logical data types. Use the DSM Options Editor to create data types and mappings. You can create multiple DSM Option objects but you can apply only one of them to the model.

You can assign data types in any of the following scenarios:

- Deriving a physical model from a logical model, or switching from a logical model to a physical model. A physical data type is automatically assigned to each column based on the data type assigned to the corresponding attribute.
- Deriving a physical model from a physical model. The corresponding data type for the target database is automatically assigned.
- Migrating from one version of a database to a newer version, or changing the target database. You are prompted you to verify if you want to convert data types. When you convert the data types, the corresponding data type for the target database is automatically assigned.
- Creating a physical model by reverse engineering from a database or script file. Information from the database or script file is extracted, and a physical data type is automatically assigned to each column.

You can apply data type standards in three ways, as illustrated in the following diagram:



To apply data type standards, follow these steps:

Decide a method to apply standards.

Applying Data Type Standards

To create and apply data type standards, follow these steps:

1. Define data type standards from Model Explorer.
2. (Optional) Create a mapping.
3. Apply data type standards to the model.

To import data type standards from a previous version, follow these steps:

1. Import data type standards.
2. Apply data type standards to the model.

Apply data type standards by binding a model template:

1. Bind a model template.

Back to Top

Decide a Method to Apply Standards

You can apply data type standards in three ways:

Create a standard

Create a data type standard and apply it to your model if you are applying standards for the first time. After you create a DSM object and define your standards, you can use the standard in all your data models.

Import from a previous release

Import data type standards from a previous release and apply the standard to your model. Importing saves your time from creating a standard and maintains the consistency across all data models in your organization.

Bind a model template

Bind an existing model template having data type standards to your model. You can use the same template for multiple models. You can also sync the template with the current model for any changes in the model. After you sync the template, it will inherit the changes that are made in the current model.

Back to Top

Create and Apply Data Type Standards

Create data type standards if you are implementing standards for the first time.

Define Data Type Standards from Model Explorer

Define data type standards to maintain uniform data type standards across all data models.

Follow these steps:

1. Expand Model Explorer, right-click DSM Options, and click New.

   A data type standards object is added.

2. Right-click the newly created object and click Properties.

   The DSM Option Editor opens.

3. (Optional) Rename the DSM object in the Name field.
4. Click the Logical Definition tab and edit the Domain, Length, Precision, and Default Length values for the required data type.
5. (Optional) To create a data type, click New under the Logical Definition tab.

   A data type is created.

6. (Optional) Rename the new data type and change the properties.

   A data type standard object is defined and saved.

> To delete a DSM object or a data type, select the DSM object or the data type row and click Delete.

If you want to create a mapping, you do not have to close the DSM Option Editor window.

Back to Top

(Optional) Create a Mapping

Map data types between different databases to maintain uniform data types in all the data models. Mapping lets you eliminate the differences in the data types when you import a model from one database to another. Also, mapping enables you to use new data types in physical models.

> You can create multiple DSM Datatype Mapping objects and DSM Datatype Mapping entries in the DSM Datatype Mapping Editor.

Follow these steps:

1. In the DSM Option Editor, select the Mapping tab and click New.

   A DSM Datatype Mapping object is created.

2. Click Edit.

   The DSM Datatype Mapping Editor opens.

3. (Optional) Edit the name of the DSM Datatype Mapping object.
4. Select the Mapping tab, and select the source database from the From drop-down list and the target database from the To drop-down list.
5. (Optional) Edit the To Datatype field.
6. (Optional) Add a data type for mapping.
7. Click Close.

   The DSM Datatype Mapping Editor closes and you return to the DSM Option Editor dialog.

Back to Top

Apply Data Type Standards to the Model

After you import the data type standards object, apply the standards by attaching the DSM object to the data model.

To attach a DSM object to a model, open the DSM Option Editor from Model Explorer, and select the Is Active check box.

Back to Top

Import Data Type Standards from a Previous Version

If you have created data type standards in a previous version of erwin DM, you can import it to erwin DM version 9. When you import data type standards, mappings are also imported.

Back to Top

Import Data Type Standards

If you have upgraded to the latest release of the r9 series, you can import the data type standards from a previous version. Importing the standards maintains consistency across all data models in your organization and saves time from creating standards.

> The file that you are importing must have the .DSM extension.

Follow these steps:

1. Expand the Model Explorer, right-click DSM Options and select New.

   A new DSM Option object is added.
2. Right-click the new object (or an existing object) and select Properties.

   The DSM Option Editor appears.
3. Click Import Other.
4. Select the .DSM file from the stored location and click Open.

   The DSM file is displayed as a new DSM Option object.

Back to Top

Apply Data Type Standards to the Model

After you import the data type standards object, apply the standards by attaching the DSM object to the data model.

To attach a DSM object to a model, open the DSM Option Editor from Model Explorer, and select the Is Active check box.

Back to Top

Apply Data Type Standards by Binding a Model Template

When you bind a model template having data type standards to the current model, data type standards are also applied.

Bind a Model Template

Binding a model template to a data model copies the non-built-in objects from the template model to the target model. When you bind a model template, the predefined model objects populate your model. Therefore, binding eliminates the need to enter objects manually or depend on reverse engineering. Changes to the model template are displayed in the models that use it, maintaining consistency of object names. Whenever you open a model with a bound model template, you can synchronize any model template changes with the objects used in the model.

Binding of model templates can occur at any time necessary, including during model creation or after the model is created.

The objects that are brought in to a model from a template fully respect modeling rules. If any conflicts occur in the model, the existing data is preserved as much as possible.

Follow these steps:

1. Click File, Model Templates, Bind Template.

   The Bind Model Template dialog opens.

2. Use the toolbar to browse for the template file locally on your computer or in Mart. If you have used the bind template feature before, use the drop-down control to select the model template to bind to the current model.

3. Select one of the following options to specify how to bind the template, and click OK.

   Load the entire contents of the template

   Specifies to load the entire contents of the template. This option binds all of the model template objects to the current model without the opportunity to review and select individual template objects.

   Run the Template Wizard to select objects for synchronization

   Specifies to open the Model Template Synchronization Wizard so that you can select the objects in the model template to synchronize with the data model.

   The model opens in the workspace with the template objects copied into it. The data type standard is applied on the model.

Back to Top

Create Reports Using Report Designer

Report Designer lets you create a wide variety of reports and on the current data available in your data models. If you are working in the Workgroup Edition, you can create Mart administrative reports and cross-model reports. You can create report templates for frequently used reports and you can export reports to HTML, CSV, and PDF formats. You can create report solutions and include multiple report templates.

You cannot reuse the .RTB reports that were created using Report Template Builder, because the metamodel objects are changed. You must recreate the reports in Report Designer.

Several sample (pinned) reports are included with the application and they are available in the following location:

```
%Installdir%\erwin\Data Modeler r9\BackupFiles\Report Solutions
```

Also at the following location:

Windows 7

```
C:\Users\<user name>\AppData\Local\erwin\Data Modeler\9.8
\Report Solutions
```

> The pinned reports are created using a Logical-Physical model. If you use the pinned reports with a Logical-only or a Physical-only model, it may result in inappropriate results.

Follow these steps:

1. Click Tools, Report Designer. The erwin Report Designer window opens.
2. To create a report solution, click Create new solution for reports on the toolbar.

   If the current report solution is not saved already, you are prompted to save it. A new blank workspace appears.
3. To create a report template, follow these steps:
   a. Click Create a Report on the toolbar. The Report Editor dialog appears.
   b. Enter a name for the report.
   c. Select the model type that you want to report on. Select Mart to create a report on the Mart data.

The items in the Select Report Subject pane change according to the Report Type that you select.

> You must have the Reports: View permission to create Mart reports.

d. In the Select Report Subject pane, select the item on which you want to generate the report.

e. In the Select Report Fields pane, select the fields that you want to include in the report.

f. To create a report that filters specific objects and includes them in the report, use the Filter by option. This option is available only for entities, tables, attributes, columns, subject areas, and indexes.

g. Click Run Report.

If you are creating a report on your Mart and you have selected an option under Filter By, the Mart Report Filter dialog opens.

h. Select the specific models that you want to get the results from. Enter the specific model object name. Use wildcard characters, if necessary.

The report is generated and displayed. If you do not specify any model or entity, the generated report includes all the models. If you have used a wildcard, by default, the wildcard is applied to all the names. For example, if you type ID, the report includes Cust_ID and ID_For_Category. If the result is very large for Mart reports, the report is displayed in multiple pages (data grid). Use the navigational icons in the toolbar to view the subsequent pages.

i. (Optional) Click the Clear data in data grid icon to clear the results in the grid.

4. To configure report preferences, follow these steps:

a. Click Edit, Preferences. The Report Designer Preferences dialog appears.

b. Complete the following fields and click OK:

Add object type as prefix for the properties 'Name' and 'Physical Name'

Prefixes the object type to only the Name and Physical Name columns. For example, you are generating a report on entities and the corresponding attribute names and data types. The column names in the report are displayed as Entity Name, Attribute Name, and Attribute Logical Data Type.

Add object type as prefix for all other properties

Prefixes the object type to all columns.

> These options work only when they are selected at the time of creating a report. If these options are selected at the time of modifying an existing report, they do not work.

Apply optimization mode if row count is more than

Specifies the number of rows at which the computer resources are optimized to display the report. If the row count in your report exceeds the number that you specify here, the report is displayed in optimization mode. In this mode, the group and sort options are not available in the result grid. If no value is entered, the row count for optimization mode is considered as equal to the total row count of a report.

Number of top rows to display

Specifies the number of top rows that you want to display. This option displays the first few rows that match a report criteria, when the result contains more rows than the number specified here.

5. To generate a report from an existing report template, select the template, right-click, and click Run.
6. To modify an existing report template, select the template, right-click and then click Properties. After you finish modifying, restart erwin DM. Your changes will take effect when you run the report again..

> When you modify an existing report template and select more properties for the existing report subject, the selections are retained. If you select a different report subject, the previous report subject and its properties are not included in the report.

7. To customize the appearance of the report at the node-level, follow these steps:
   a. Right-click the node and click Properties. Complete the following fields:

Name

Specifies the name of the node. By default the name of the selected object is displayed.

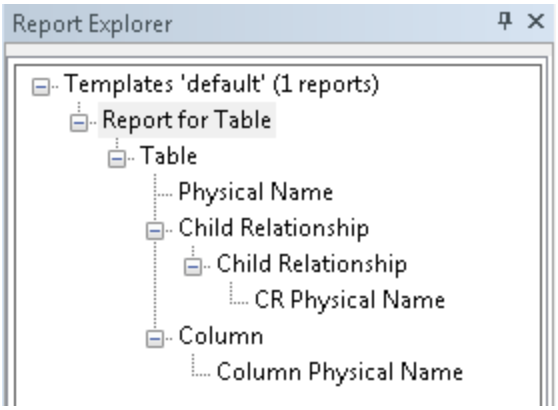
HTML Style

> These settings apply to both HTML and PDF reports.

Font Family: Specifies the font for the selected node.

Size: Specifies the font size for the selected node.

PDF Page Breaking: Specifies that you want to display the details of each node on a new page. The node for which this option is selected and the subsequent nodes are displayed on a new page. This option is not effective when the report has only one node. When the report has multiple nodes, do not set this option on the main node. Suppose your report nodes look like the following illustration:



If you apply the PDF Page Breaking option to the Child Relationship (above CR Physical Name) node, the CR Physical Name and Column Physical Name nodes are displayed on a new page.

Foreground: Specifies the text color.

Background: Specifies the background color.

Template Color: Specifies that the color that is specified in the template is used.

Bold/Italic/Underline/Strikeout: Specifies the effect for the text.

Width: Specifies the width of the selected node (report column). You can specify the width for any node and it is applied when the report is exported to HTML or PDF. However, the width is not applied to the last node (last column in the report) in a PDF report. This is because, the remaining length of the fixed page width is considered as the actual width of the column.

Alignment: Specifies the alignment of the text.

8. To export a report solution to HTML, follow these steps:
   - Right-click the report solution and click Export All to HTML. The Report Export dialog appears.
   - Click Browse and select the folder to which you want to export the report.

- Go to the HTML tab and add the title in the Solution Title in HTML field.
- Click OK. The report is generated in the specified folder.

9. To export a report to HTML, follow these steps:
   a. Click the Clear data in data grid icon and clear the result grid.

      This is a best practice and helps the application export the report without consuming much of the computer resources.

   b. Right-click the report and click Export to HTML. The Report Export dialog appears.
   c. Click Browse and select the folder to which you want to export the report.
   d. Go to the HTML tab and complete the following fields:

      Select the template for exporting to HTML

      Specifies the templates that you want to use with the HTML report. You can use customized templates, provided the formatting is similar to the default templates. However, if you modify HTML templates, you must restart erwin DM for the changes to take effect.
      The default templates are available in the following folder:

      C:\Program Files\erwin\Data Modeler r9\Report\Template\Template-Default

      Solution Title in HTML

      Specifies the title for the entire report solution.

      Report Header in HTML

      Specifies the text that appears between the report title and the first line of the report.

      Report Footer in HTML

      Specifies the text that appears at the bottom of the report.

   e. Click OK. The report is generated in the specified folder. If the report data spans multiple page grids in the Report Designer, the data is segregated into multiple sections in the HTML. The sections are first divided by model name and then by object name. This means that a section may include objects from different models, which are sorted by model name and object name.

10. To view the HTML report on a different computer, include the computer name in the path to the filename. For example, if you have published the report to c:\erwin\reports folder on a computer that is named Comp001, use the following path to access it:

```
\\comp001\erwin\reports\index.htm
```

> The computer from which you are accessing the report must be in the same domain as the one in which the report is published. In addition, the user who is accessing the report must have at least the read-only privileges on the corresponding folder.

11. To export a report or a report solution to CSV, follow these steps:
    a. Right-click the report and click Export to CSV. Or, right-click the report solution and click Export All to CSV. The Report Export dialog appears.
    b. Click Browse and select the folder to which you want to export the report.
    c. Go to the TEXT tab and select the following options:

       Format

       Specifies whether you want to include the first-level object names multiple times. For example, you have included entity names and attribute names in your report and you have selected No Duplicates. The entity name appears only once for all the corresponding attribute names. If you have selected Duplicates, the entity name appears as many times as the corresponding attribute names appear.

       Text Qualifier

       Specifies the symbol that you want to use to indicate the column headers.

       Delimiters

       Specifies how you want to separate two columns in the exported report. The option Comma Delimited creates a .CSV file and the remaining options create .TXT files.

    d. Click OK. The report is generated in the specified folder.
12. To export a report to PDF, follow these steps:
    a. Click the Clear data in data grid icon and clear the result grid.

       This is a best practice and helps the application export the report without consuming much of the computer resources.

    b. Right-click the report and click Export to PDF. The Report Export dialog appears.
    c. Click Browse and select the folder to which you want to export the report.

d. Go to the PDF tab and enter values/set the following fields:

Solution Title in PDF

Specifies the title for the entire report solution.

Report Header in PDF

Specifies the text that appears on the first line of every page of the report.

Report Footer in PDF

Specifies the text that appears at the bottom of every page of the report.

For each of these options, you can specify the font, size, color, and alignment.

Image Orientation

Specifies the direction of the images that are generated in the report. Images are generated for the Subject Area and ER Diagram reports, when you select properties under Graphical Members.

Font & Color

Specifies the styles to be used with respect to the font and its color.

Table of Contents

Indicates whether the report should contain a table of contents.

Index

Indicates whether the report should contain an index.

Comprehensive Report

Indicates whether the report should contain comprehensive tabular information based on the items selected for the report. Selecting this option lets you avoid information duplication in the report.

e. Go to the Advanced tab and enter values/set the following fields:

Report Logo

Lets you add a logo to the report. Also, lets you link the logo to a URL of your choice.

Logo Position

Specifies the position of the logo in the report.

Alignment

Specifies the alignment of the logo at the selected position in the report.

Custom Report

Lets you set custom colors to the various sections of the report. Select the Use Custom Colors check box to be able to select colors of your choice.

Select ER Diagram Format

Specifies the format of the ER diagram.

Page Size

Specifies the page size that would be used for printing the PDF copy of the report.

f. Click OK. The report is generated in the specified folder. If the report data spans multiple page grids in the Report Designer, the data is segregated into multiple sections in the PDF. The sections are first divided by model name and then by object name. This means that a section may include objects from different models, which are sorted by model name and object name.

13. To add a report from another report solution, follow these steps:
    - In the Report Designer dialog, right-click the template name and click Add Report.
    - Select the report that you want to include (.erpt file) and click OK.
    - Save the report solution.

This section contains the following topics

[Model Validation Reports](#)
[Pinned Reports from the Report Designer Menu](#)
[The Filter By Option](#)
[Report Designer Features](#)
[Subject Area Reports](#)
[Change the Logo on the HTML Report](#)
[Generating TLX Reports](#)
[Dimensional Modeling](#)

Model Validation Reports

You can now generate model validation reports such as tables that do not have columns and columns that do not have comments. The reports can be generated using filters, and filters are

available for the following objects:

- Entity/Table
- Attribute/Column
- Subject Area
- Relationship
- Key group/Index
- Domain

Follow these steps:

1. Open erwin Report Designer and create a report.
2. Select Logical, Physical, or Logical/Physical. For example, select Logical.
3. Select a subject in the Select Report Subject pane. For example, select Entity.
4. In the Select Report Fields pane, expand the Filter node for the required object and select the required criterion. For example, expand Entity, Filter, and select No Definition. Expand Properties and select Name.
5. Click Run Report.The report opens in erwin Report Designer.

   In this example, the report includes the names of entities that do not have a definition.

Pinned Reports from the Report Designer Menu

You can now open the built-in reports from the Open pinned reports from disk toolbar item. The pinned reports are available in the local computer of the user. These reports are segregated into categories and each category of reports is available in a separate folder. You can create a folder of your own add the reports of your choice to it.

The following folder includes the built-in reports:

Windows 7

```
C:\Users\<user name>\AppData\Local\erwin\Data Modeler\9.8\Report Solutions
```

Reports that are copied directly under this folder do not appear in the Pinned Reports dialog. The reports must be in a subfolder under this folder.

Follow these steps:

1. Launch erwin Report Designer and click Open pinned reports from disk on the toolbar.

   The Pinned Reports dialog appears.
2. Select the category of the report that you want to see.
3. Select one or more reports that you want to run and click OK.

   The report is included to the current Report Solution and the Pinned Reports dialog closes.

The Filter By Option

Report Designer includes the *Filter By* option that lets you generate drill-down reports. Some of the drill-down reports that you can generate are as follows:

Entity/Table level reports

- Entities and tables in the current Subject Area
- Entities and tables that are selected in the current diagram

Attribute/Column level reports

- Migrated attributes and columns
- Owned attributes and columns
- Role-named attributes and columns
- Foreign (FK) Key groups and indexes
- Generated indexes
- Nonunique (IE) Key groups and indexes
- Primary (PK) Key groups and indexes
- Unique (AK) Key groups and indexes

Subject Area reports:

- All entities and tables in the current subject area
- Members of the current diagram
- Members of all the diagrams in the current Subject Area
- Members of all the diagrams in the Main Subject Area, that is, the members of all diagrams at the model level
- Members of the diagrams that are specified at run time

You can use the Graphical Member option for diagrams and can view the images of diagrams.

The filters for Entity/Table and Key Group/Index are available when you select the members of a diagram or a Subject Area.

Attributes and columns can be displayed the following order:

- Alphabetic
- Attribute (the order in which attributes appear on the logical model)
- Column (the order in which columns appear on the physical model)
- Physical (the order in which columns appear in the DDL after Forward Engineering)
- Reverse Alphabetic

Report Designer Features

Report Designer includes the following features:

Count Objects

Based on the Report Type selection on the Report Editor, it displays the count (number) of Subject Areas, Entities/Tables, Attributes/Columns, Key groups/Indexes, and Relationsips that are present in the model. You can export this count to CSV format. On the Report Designer, click Export > Export Count to CSV.

Report Explorer Features

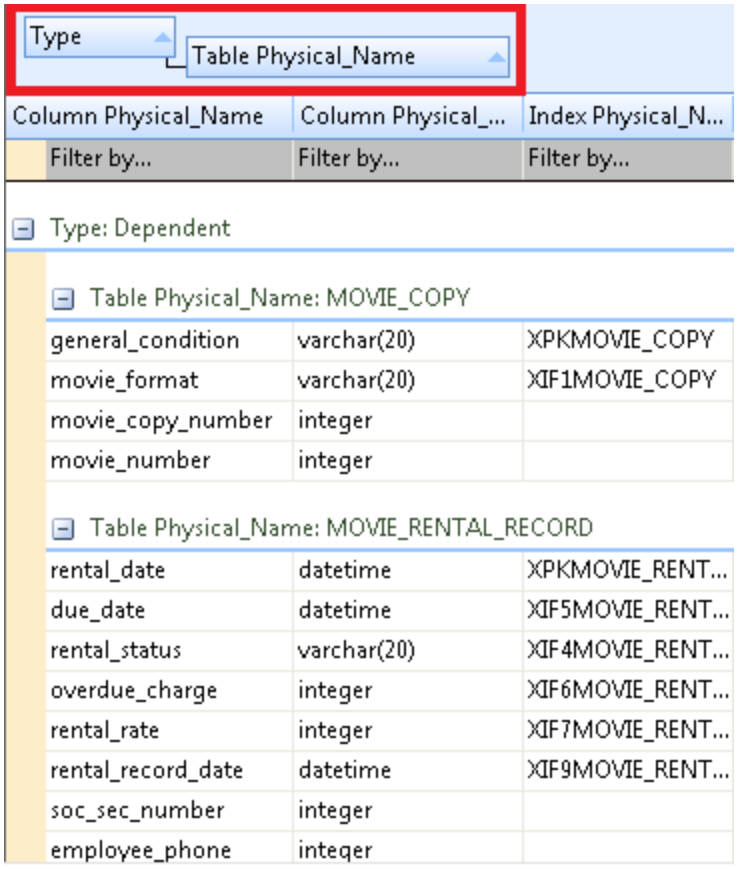
Move child nodes: To move the child nodes up or down within a parent node, right-click the node and click Move Up or Move Down. This option is to move child nodes within a parent node and not to move parent nodes in a report.

Result Grid Features

- Drag-and-drop: To change the order of columns, drag-and-drop the report node to the desired position in the Report Explorer pane. Select the report template and click Run current report on the toolbar.
- Sort: To sort the report results on a specific column, click the column header. The default sort order is the order in which you have created the objects in the model.
- Hide a column: To hide a column, click the column header and drag it away from the report until you see the cross mark. To display a hidden column, right-click a column header, click Columns, and then select the column that you want to display.

  > Hidden columns are included when you export a report to CSV or HTML.

- Group the report result by a column: To group the report result by a column, click the column header and drag it to the *Drag a column header here to group by that column* section. The following illustration shows a report that is grouped by two columns:



- Rename a column: To rename a column, right-click the corresponding node in the Report Explorer and click Properties. Enter a new name in the Node Name field and click OK. Select the report template and click Run current report on the toolbar.
- Configure the look and feel of each column: To configure the look and feel of each column, right-click the corresponding node in the Report Explorer and click Properties. Complete the fields in the Customize report style section. Be aware that the styles applied to a parent node are *not* applied to child nodes automatically.
- Clear grid: To clear the data displayed in the result grid, click View, Clear gird. Clearing the result grid is a best practice when exporting a large or a complex report to HTML.
- Export to HTML, TXT, and PDF: To export a report or a Report Solution, right-click it and click Export to HTML, Export to CSV, or Export to PDF. When you export a report to CSV, select the delimiter as Comma Delimited to view the report in CSV format.

Pinned Reports

You can open the built-in reports from the Open pinned reports from disk toolbar item. The pinned reports are available in the local computer of the user. The reports are segregated into categories and each category of reports is available in a separate folder. You can create a folder of your own add the reports of your choice to it.

Several sample (pinned) reports are included with the application and they are available in the following location:

```
%Installdir%\erwin\Data Modeler r9\BackupFiles\Report Solutions
```

Also at the following location:

Windows 7

```
C:\Users\<user name>\AppData\Local\erwin\Data Model-
er\9.8\Report Solutions
```

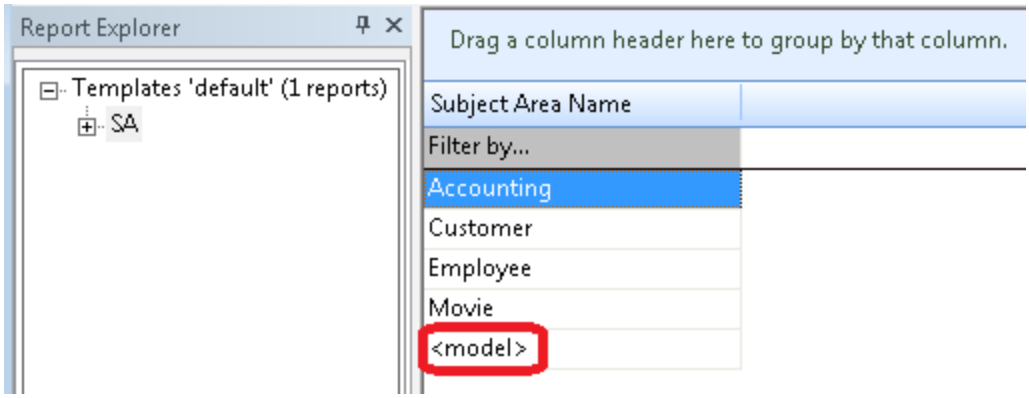
> The pinned reports are created using a Logical-Physical model. If you use the pinned reports with a Logical-only or a Physical-only model, it may result in inappropriate results.

Reports that are copied directly under this folder do not appear in the Pinned Reports dialog. The reports must be in a subfolder under this folder.

Subject Area Reports

By design, a model is considered as the main Subject Area. Therefore, when you create a generic report on Subject Area and include its properties such as Name, the model is also displayed in the list of Subject Areas. However, the model name is not included when you generate a report on specific Subject Areas. You can generate a report on specific Subject Areas by selecting the *Specify Diagrams at Run Time* option.

The following diagram shows a generic report of Subject Area:



In the previous report, <model> is the model and the remaining entries are Subject Areas.

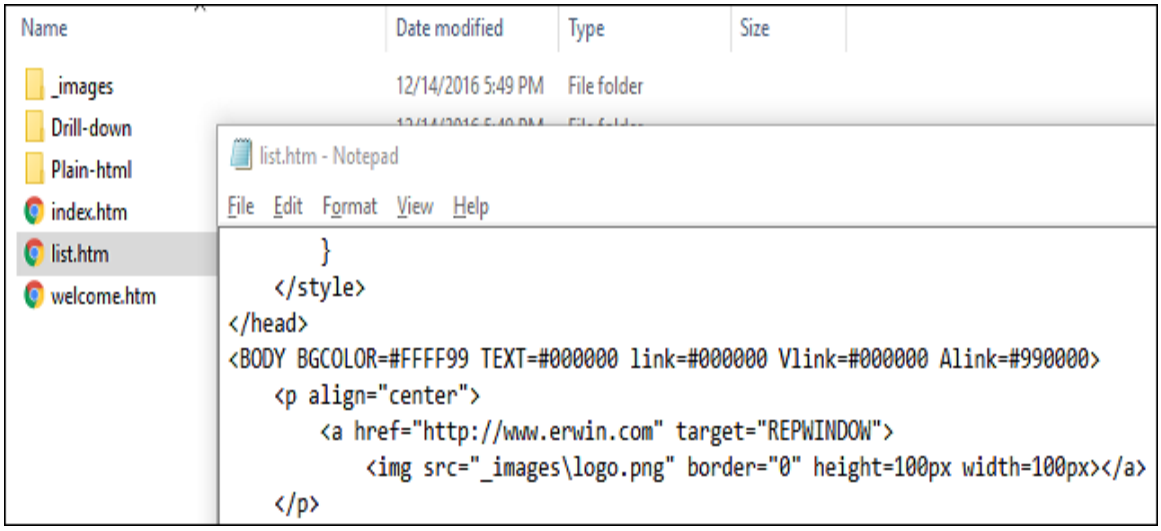
> If you generate a Subject Area report for a Mart model, <model> is not included as a Subject Area. To generate a Subject Area report, we recommend that you open the Mart model and generate the report as a "local" model.

Change the Logo on the HTML Report

By default, erwin logo is displayed on HTML reports and the logo is linked to the erwin website. You can change the logo and link it to a website of your choice.

Follow these steps:

1. Open the folder where erwin® Data Modeler is installed.
2. Navigate to the \Report\Template\Template-Default_images folder.
3. Replace the image in the logo.png file.
4. Navigate to the \Report\Template\Template-Default folder.
5. Open the list.htm file in an HTML editor.
6. Replace http://www.erwin.com with the required URL, as shown in the following image:



7. Restart the application.

   The logo and the link to the website are changed when you generate a report the next time.

Generating TLX Reports

A TLX report is a model validation report that uses erwin Template Language (TLX). Report Designer includes some default validation reports and also lets you configure new reports with custom properties. You must be familiar with erwin Template Language (TLX) to configure new reports.

When you configure new reports, you can add two types of custom properties, filters and TLX properties. The code for a filter includes a function that returns TRUE or FALSE.

Generate a Default TLX Report

Report Designer includes default TLX reports at entity/table, attribute/column, domain, and relationship level. Some of the properties that are included in the default TLX reports are no definition or no primary key. This topic describes how you can create an entity/table level report as an example. You can create other TLX reports on the same lines as described in the procedure.

To create an entity/table level report, follow these steps:

1. Click Tools, Report Designer, Create a new report.
2. Select Logical/Physical for Report Type.
3. Click Entity/Table in the Select Report Subject pane.
4. In the Select Report Fields pane, expand Filter By and select the No Definition and No Comment check boxes.
5. Expand Properties and select Name, Physical Name check boxes.
6. Click Run Report.

   The list of entities and tables that do not have a definition or a comment is displayed.

Create a Custom TLX Report

If you are familiar with TLX scripting, you can add your own functions for properties of your choice and configure custom reports. For example, you can create a report that gives the number of attributes in each entity. You can add two types of custom properties, filters and TLX properties. The code for a filter includes a function that returns TRUE or FALSE.

To configure a new report, you must modify the following configuration files:

- TLXReportAssembly.xml
- TLXReportTemplate.FET

These files are available in the <erwin install folder>\Report\TLX folder.

Follow these steps to create a custom TLX report:

1. Modify the TLXReportAssembly.xml file and include the TLX property that you want to report on.
2. Modify the TLXReportTemplate.FET file and include the function for the custom property.
3. Restart erwin Data Modeler.
4. Create a report in the Report Designer and select the TLX property that you have added.
5. Run the report.

Modify the TLXReportAssembly File

To display a property under the Filter By node or the Property node in the Report Designer, you must add an entry for the property in the TLXReportAssembly file.

The TLXReportAssembly.xml file includes the following parameters:

Assembly ID

Specifies the unique identifier for the property entry in the TLXReportAssembly file. You can enter any number of your choice, as long as it is unique within this file.

OwnerID

Specifies the owner ID for the object for which you want to create a custom property. Use the following owner IDs:

Owner ID	Object
1075838979	Entity/Table
1075838981	Attribute/Column
1075838983	Domain
1075839016	Relationship
1075838985	Key Group

Owner ID	Object
1075838978	Custom Template
1075839014	Subject Area

Name

Specifies the name of the custom property the way it appears in the Report Designer.

ExtractType

Specifies the type of report for which the custom property is available. 1 denotes Logical, 2 denotes Physical, and 3 denotes Logical/Physical.

DBMS Info

Specifies the database for which the custom property applies. Enter the information in the format, "database brand code, major version number of the database, minor version of the database."

A value of "0,0,0" indicates that the custom property applies to all databases. The following table provides the DBMS information for the databases that the current release supports:

Database	DBMS Info in TLXAssembly
AlloyDB	(1075918980, 1, 1)
ArangoDB	(1075859217, 3, 0)
Apache Avro 1.x	(1075859205, 1, 0)
Amazon Keyspaces	(1075859223, 2, 0)
Azure Synapse SQL Pool	(1075859211, 10, 0)
Azure Synapse Fabric Lakehouse	(1075859211, 11, 0)
Cassandra	(1075859199, 3, 0)
Couchbase 6.x	(1075859202, 6, 0)
Couchbase 7.x	(1075859202, 7, 0)

Database	DBMS Info in TLXAssembly
Databricks	(1075859232, 1, 0)
Db2 for i 5.x/6.x/7.x	(1075859019,4,0)
Db2 for LUW 9.5	(1075858977,9,5)
Db2 for LUW 9.7/10.x	(1075858977,9,7)
Db2 for LUW 11.1/11.5	(1075858977,11,1)
Db2 for z/OS 8.1	(1075858978,8,1)
Db2 for z/OS 9.1/10/11	(1075858978,9,1)
Db2 for z/OS 12	(1075858978,12,0)
DynamoDB	(1075859229,19,0)
Google BigQuery	(1075859226,2,0)
Hive 2.1.x	(1075859187,2,1)
Informix 10.x/11.x/12.x	(1075859006,9,2)
JSON 1.x	(1075859208,1,0)
MongoDB 4.x	(1075859196,4,0)
MongoDB 5.x	(1075859196,5,0)
MongoDB 6.x/7.x/8.x	(1075859196,6,0)
MariaDB 10.4.x	(1075859190,10,4)
MySQL 8.x	(1075859129,8,0)
Neo4j 4.2.x	(1075859214, 4, 2)
Neo4j 4.3.x/4.4.x	(1075859214, 4, 3)

Database	DBMS Info in TLXAssembly
Netezza	(1075918978,7,2)
ODBC/Generic 2.x	(1075859009,2,0)
ODBC/Generic 3.x	(1075859009,3,0)
OpenAPI	(1075885345, 3, 0)
Oracle 12c r2	(1075858979, 12, 0)
Oracle 18c	(1075858979, 18, 0)
Oracle 19c	(1075858979, 19, 0)
Oracle 21c	(1075858979, 21, 0)
Parquet 2.x	(1075859220, 2, 0)
PostgreSQL 12.x/13.x/14.x/15.x	(1075859235,12,0)
PostgreSQL 16.x	(1075859235,16,0)
Progress 9.x/10.x/11.x	(1075859010,9,0)
Redshift 1.0	(1075918979,1,0)
SAS	(1075859013,1,0)
SAP ASE 15.x/16	(1075859017,12,5)
SAP IQ 15.x/16	(1075859130,15,0)
SQL Server 2012	(1075859016,11,0)
SQL Server 2014	(1075859016,12,0)
SQL Server 2016/2017	(1075859016,13,0)
SQL Server 2019/2022	(1075859016,15,0)

Database	DBMS Info in TLXAssembly
SQL Azure	(1075859016,16,0)
Snowflake	(1075859193,4,10)
Teradata 14.10	(1075859018,14,10)
Teradata 15.x/16.x	(1075859018,15,0)
Teradata 17.x/20.x	(1075859018,17,0)

TLXFunction

Specifies the TLX function name for the custom property. The function name can be any name that describes the property.

Is Filter

Specifies whether the custom property is a filter or a property. Include a value of 1 for filter and 0 for property. If you include 1, the property is displayed under the Filter by node of the corresponding object. If you include 0, the property is displayed under the Properties node of the object in Report Designer.

Follow these steps:

1. Open the TLXReportAssembly.xml file in a text editor such as Notepad.
2. Add a line for the property that you want to include in the report.
3. Save the file.

Modify the TLXReportTemplate.FET File

The TLXReportTemplate.FET file includes the functions that help the Report Designer generate the report for a TLX property. You must modify the file to include the function for a custom property for which you want to generate a report.

> Use the same function name in the TLXReportAssembly and TLXReportTemplate files.

Follow these steps:

1. Open the TLXReportTemplate.FET file in a text editor such as Notepad.
2. Add the TLX script for the property that you want to include in the report.

3. Save the file.

   Now the custom property appears in the Report Designer under the corresponding node. You can include the property in a report.

Example--Number of Attributes in an Entity

This example demonstrates how you can generate a report on the custom property *Number of Attributes in an Entity*. This property is of type TLX property.

Follow these steps:

1. Add an entry to the TLXReportAssembly.xml file to create a node for Number of Attributes in Report Designer.
2. Add the TLX function NumberofAttributes to the TLXReportTemplate.FET file.
3. Create a report and include the Number of Attributes property.

Add an Entry to the TLXReportAssembly.xml File

Follow these steps:

1. Open the TLXReportAssembly.xml file from the <erwin install folder>\Report\TLX folder.
2. Add the following entry and save the file:

```
<Assembly Id="1002"  OwnerId="1075838979" Name="Number of Attributes"
ExtractType="1" DBMSInfo="0, 0, 0" TLXFunction="NumberofAttribute"
IsFilter="0"/>
```

Add the TLX Function NumberofAttributes to the TLXReportTemplate.FET File

Follow these steps:

Add the following code to the TLXReportTemplate.FET file and save the file:

```
SPItemBegin = NumberofAttribute

[

        Set("Count", 0)

        ForEachOwnee("Attribute")

        {

                Increment("Count")
```

```
        }

        Value("Count")

]

SPItemEnd
```

Number of Attributes is added to Report Designer under Entity, Properties.

Create a Report and Include the Number of Attributes Property

Follow these steps:

1. Open Report Designer.
2. Select the Report Type as Logical.
3. In the Select Report Subject pane, select Entity.
4. In the Select Report Fields pane, expand Entity, Properties, and select Name, Number of Attributes.
5. Click Run Report.

Dimensional Modeling

The ability to produce reports on data source mapping is added to Report Designer in erwin® Data Modeler Release 9.5.2. In the data warehouse environment, it is important to track the following information:

- Source of the data.
- The method by which the source data is extracted and cleansed before it is imported into the data warehouse.
- The frequency and timing of data warehouse updates.

You may have a data warehouse that combines information from several online transaction processing (OLTP) systems and archived data into a single decision support system. Your data may also come from relational and non-relational sources. To support regular updates and data quality checks, you must know the source for each column in your data warehouse. You must also know when and how the data is updated. You can document all of this information using erwin® Data Modeler. This information can also be produced as reports in various formats using Report Designer.

Report on Data Sources, Target Columns, and Transformations

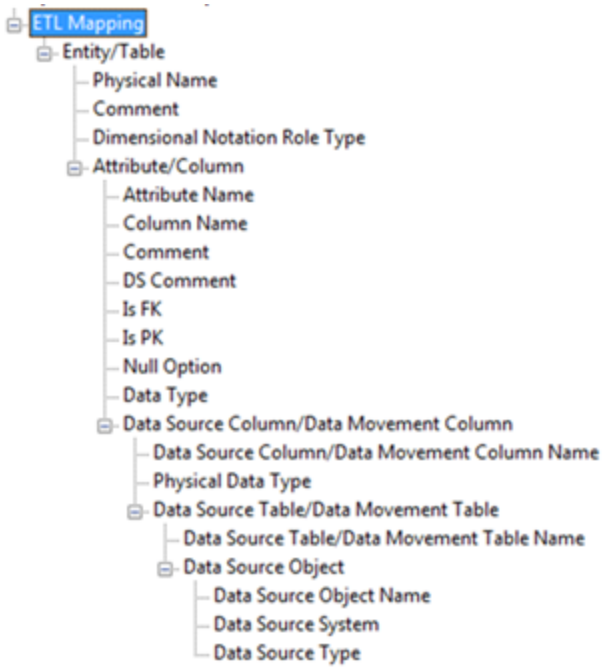
This report includes the list of data sources, target columns, and the corresponding transformation details.

> To generate this report, you must have a model that documents data movement sources and has the column level mappings defined.

Follow these steps:

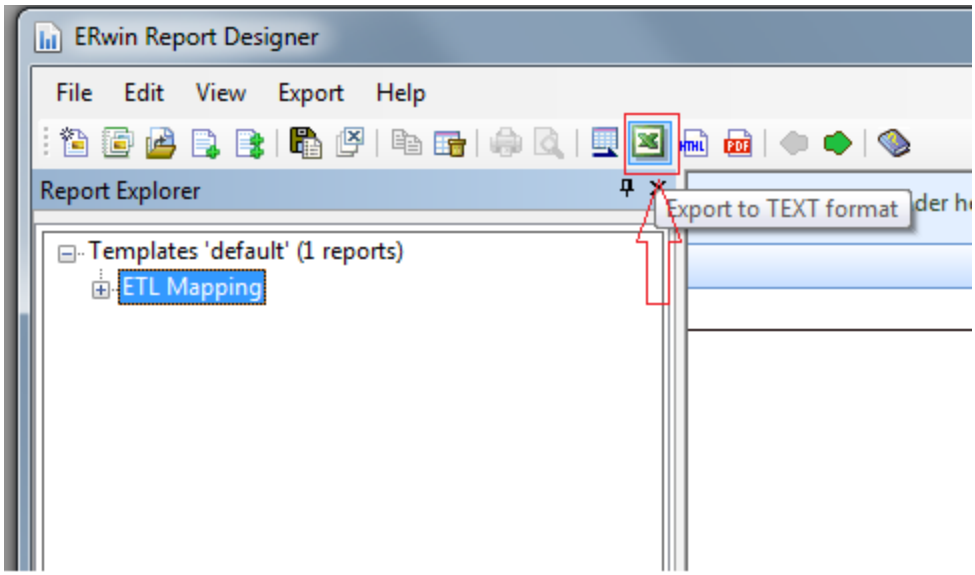
1. Open erwin Data Modeler and open a model in which data movement sources and column level mappings are defined.
2. Click Tools, Report Designer.
3. Click Create New Report. The erwin Report Editor dialog opens.
4. Enter the title of the report as ETL Mapping.
5. Enter a name for the report and select the Report Type as Logical/Physical.
6. Select the Report Subject as Entity/Table.
7. In the Select Report Fields pane, expand Entity, Properties, and select Name, Definition.
8. Expand Entity/Table, Properties, and select Comment, Dimensional Notation Role Type, and Physical Name.
9. Expand Attribute/Column, Properties, and select Comment, DS Comment, Name, Null Option, and Physical Data Type.
10. Expand Data Source Column/Data Movement Column, Properties, and select Name, Physical Data Type.
11. Expand Data Source Table/Data Movement Table, and select Name.
12. Expand Data Source Object, Properties, and select Name, Data Source System, Data Source Type.
13. Expand Attribute, Key Type and select Is PK, Is FK.

14. Cick OK. The report template is added to erwin Report Designer.

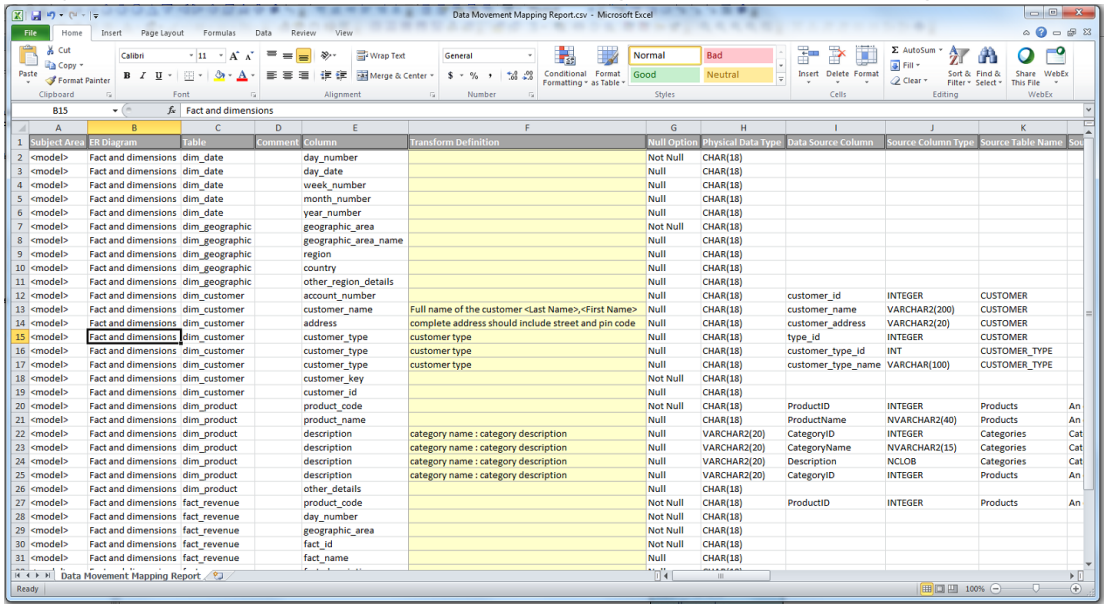


15. In erwin Report Designer, right-click the template and click Export to TEXT.

    The report is generated and displayed.



16. Verify that the report contains data source mapping information for each target column.



Data Movement Mapping Report.csv - Microsoft Excel

Subject Area	ER Diagram	Table	Comment	Column	Transform Definition	Null Option	Physical Data Type	Data Source Column	Source Column Type	Source Table Name	Sou
<model>	Fact and dimensions	dim_date		day_number		Not Null	CHAR(18)				
<model>	Fact and dimensions	dim_date		day_date		Null	CHAR(18)				
<model>	Fact and dimensions	dim_date		week_number		Null	CHAR(18)				
<model>	Fact and dimensions	dim_date		month_number		Null	CHAR(18)				
<model>	Fact and dimensions	dim_date		year_number		Null	CHAR(18)				
<model>	Fact and dimensions	dim_geographic		geographic_area		Not Null	CHAR(18)				
<model>	Fact and dimensions	dim_geographic		geographic_area_name		Null	CHAR(18)				
<model>	Fact and dimensions	dim_geographic		region		Null	CHAR(18)				
<model>	Fact and dimensions	dim_geographic		country		Null	CHAR(18)				
<model>	Fact and dimensions	dim_geographic		other_region_details		Null	CHAR(18)				
<model>	Fact and dimensions	dim_customer		account_number		Null	CHAR(18)	customer_id	INTEGER	CUSTOMER	
<model>	Fact and dimensions	dim_customer		customer_name	Full name of the customer <Last Name>,<First Name>	Null	CHAR(18)	customer_name	VARCHAR2(200)	CUSTOMER	
<model>	Fact and dimensions	dim_customer		address	complete address should include street and pin code	Null	CHAR(18)	customer_address	VARCHAR2(20)	CUSTOMER	
<model>	Fact and dimensions	dim_customer		customer_type	customer type	Null	CHAR(18)	type_id	INTEGER	CUSTOMER	
<model>	Fact and dimensions	dim_customer		customer_type	customer type	Null	CHAR(18)	customer_type_id	INT	CUSTOMER_TYPE	
<model>	Fact and dimensions	dim_customer		customer_type	customer type	Null	CHAR(18)	customer_type_name	VARCHAR(100)	CUSTOMER_TYPE	
<model>	Fact and dimensions	dim_customer		customer_key		Not Null	CHAR(18)				
<model>	Fact and dimensions	dim_customer		customer_id		Null	CHAR(18)				
<model>	Fact and dimensions	dim_product		product_code		Not Null	CHAR(18)	ProductID	INTEGER	Products	An
<model>	Fact and dimensions	dim_product		product_name		Null	CHAR(18)	ProductName	NVARCHAR2(40)	Products	An
<model>	Fact and dimensions	dim_product		description	category name : category description	Null	VARCHAR2(20)	CategoryID	INTEGER	Categories	Cat
<model>	Fact and dimensions	dim_product		description	category name : category description	Null	VARCHAR2(20)	CategoryName	NVARCHAR2(15)	Categories	Cat
<model>	Fact and dimensions	dim_product		description	category name : category description	Null	VARCHAR2(20)	Description	NCLOB	Categories	Cat
<model>	Fact and dimensions	dim_product		description	category name : category description	Null	VARCHAR2(20)	CategoryID	INTEGER	Products	An
<model>	Fact and dimensions	dim_product		other_details		Null	CHAR(18)				
<model>	Fact and dimensions	fact_revenue		product_code		Not Null	CHAR(18)	ProductID	INTEGER	Products	An
<model>	Fact and dimensions	fact_revenue		day_number		Not Null	CHAR(18)				
<model>	Fact and dimensions	fact_revenue		geographic_area		Not Null	CHAR(18)				
<model>	Fact and dimensions	fact_revenue		fact_id		Not Null	CHAR(18)				
<model>	Fact and dimensions	fact_revenue		fact_name		Null	CHAR(18)				

Report on Data Lineage with Diagrams

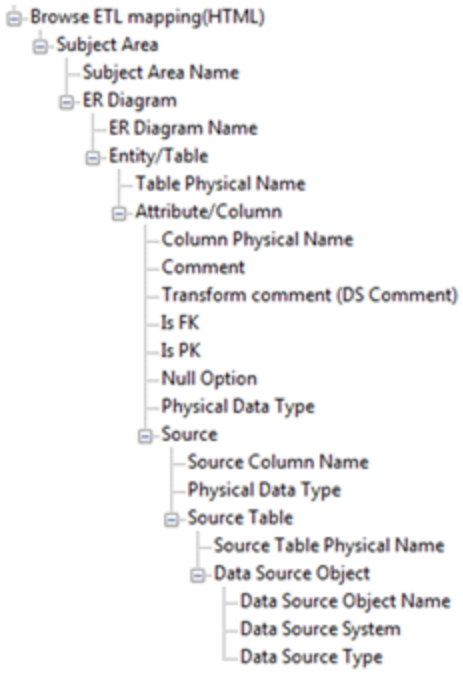
This report includes the data lineage (data source mapping) with diagrams.

> To generate this report, you must have a model that documents data movement sources and has the column level mappings defined.

Follow these steps:

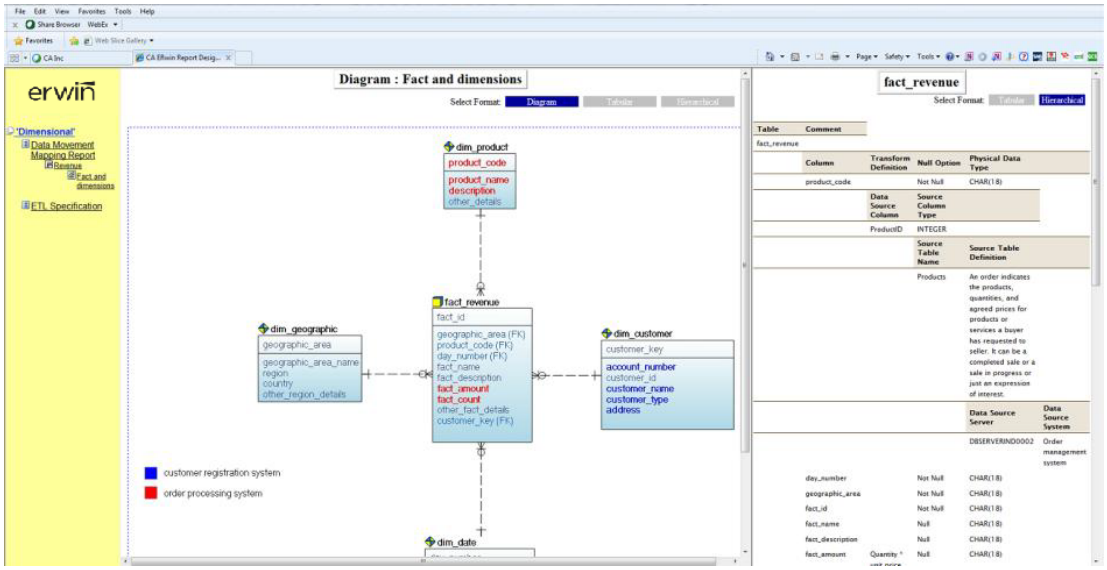
1. Open erwin Data Modeler and open a model in which data movement sources and column level mappings are defined.
2. Click Tools, Report Designer.
3. Click Create New Report. The erwin Report Editor dialog opens.
4. Enter the title of the report as Browse ETL Mapping.
5. Enter a name for the report and select the Report Type as Logical/Physical.
6. Select the Report Subject as Entity/Table.
7. In the Select Report Fields pane, expand Entity, Properties, and select Name, Definition.
8. Expand Subject Area, Properties, and select Name.
9. Expand ER Diagram, Properties, and select Name.
10. Expand Members, Table/Entity, Properties, and select Table Physical Name.
11. Expand Attribute/Column, Properties, and select Comment, Physical Name, Null Option, Physical Data Type, DS Comment.
12. Expand Data Movement Column, Properties, and select Name, Physical Data Type.
13. Expand Data Source Table/Data Movement Table, and select Name.
14. Expand Data Source Object, Properties, and select Name, Data Source System, Data Source Type.
15. Expand Attribute, Key Type and select Is PK, Is FK.

16. Cick OK. The report template is added to erwin Report Designer.



17. In erwin Report Designer, right-click the template and click Export to HTML.

    The report is generated and displayed.



18. Open the HTML report and verify the data mappings. Verify that the report contains diagrams and clickable entities through which you can navigate to the data source mapping information.

Click here to view a video demonstration of this report.

Supported Metadata Integration Bridges

The following table lists the Meta Integration Technology, Inc. (11.0.0) bridges that are currently available in erwin Data Modeler (as of Release 15.0):

> Read the information displayed in the tip text area on the Source and Destination pages of the Import from External Format and Export to External Format dialogs carefully. Complete any requirements as appropriate before proceeding.

Application	Mode
Actian Vector (Vectorwise) Database (via JDBC)	Import
Altova XMLSpy	Export
Alteryx - Unsupported Beta Bridge	Import
Alteryx Repository - Unsupported Beta Bridge	Import
Amazon Web Services (AWS) Athena / Generic Database (via JDBC)	Import
Amazon Web Services (AWS) Athena Database (via JDBC)	Import
Amazon Web Services (AWS) Aurora / MySQL Database (via JDBC)	Import
Amazon Web Services (AWS) Aurora / PostgreSQL Database (via JDBC)	Import
Amazon Web Services (AWS) RedShift Database (via JDBC)	Import
Amazon Web Services (AWS) Simple Storage Service (S3) File System	Import
Apache Avro File (JSON)	Import
Apache Cassandra NoSQL Database	Import
Apache CouchDB NoSQL Database	Import

Application	Mode
Apache Druid Database - New Beta Bridge	Import
Apache Hadoop HBase NoSQL Database (Java API)	Import
Apache Hadoop Hive Database (HCatalog and Metastore via JDBC)	Import
Apache Hadoop Hive Database SQL DDL Script	Import
Apache Kafka File System (API and Schema Registry)	Import
Apache ORC File	Import
Apache Parquet File	Import
Borland Together (via UML 2.x XMI)	Import/Export
CA Component Modeler 3.52 (ParadigmPlus)	Import/Export
CA Component Modeler 4.x (via UML 1.x XMI)	Import/Export
CA COOL:Biz 5.1	Import
CA COOL:BusinessTeam (GroundWorks) 2.2.1	Import
CA COOL:DBA (Terrain for Db2) 5.3.2	Import
CA COOL:Enterprise (ADW) 2.7	Import
CA COOL:Xtras Mapper (TerrainMap for Db2)	Import
CA Gen	Import/Export
Cloudera Data Platform Hadoop Hive Database (HCatalog and Metastore via JDBC)	Import
Cloudera Data Platform Hadoop Hive Database SQL DDL Script	Import
Cloudera Hortonworks Data Platform Hadoop Hive Database (HCatalog and Metastore via JDBC)	Import

Application	Mode
Cloudera Hortonworks Data Platform Hadoop Hive Database SQL DDL Script	Import
Cloudera Impala Hadoop Hive Database SQL DDL Script	Import
Cloudera Impala Hadoop Hive Server	Import
COBOL Copybook File	Import
Confluent Kafka File System (API and Schema Registry)	Import
Database (via JDBC)	Import
Database SQL DDL Script	Import
Databricks Delta Lake Hadoop Hive Database (HCatalog and Metastore via JDBC)	Import
Databricks Delta Lake Hadoop Hive Database SQL DDL Script	Import
DataStax Enterprise (Apache Cassandra) NoSQL Database	Import
Delimited File (CSV)	Import
Denodo Virtual DataPort	Import
EMC ProActivity 3.x & 4.0	Import
erwin 3.x (ERX)	Import
erwin 4.x Data Modeler	Import
erwin Web Portal - Vendor Internal Bridge	Export
Excel File (XLSX)	Import
File System (Delimited, Fixed Width, Excel, XML, JSON, Avro, Parquet, ORC, COBOL Copybook)	Import
Fixed Width File	Import

Application	Mode
Gentleware Poseidon (via UML 1.x XMI)	Import/Export
Google Cloud BigQuery Database (via JDBC)	Import
Google Cloud Platform (GCP) Hadoop Hive Database (HCatalog and Metastore via JDBC)	Import
Google Cloud Platform (GCP) Hadoop Hive Database SQL DDL Script	Import
Google Cloud Spanner Database (via JDBC)	Import
Google Cloud Storage (GCS) File System	Import
Grandite Silverrun-RDM 2.4.4 to 2.8.6	Import
HPE Ezmeral Data Fabric (MapR) Hadoop Hive Database (HCatalog and Metastore via JDBC)	Import
HPE Ezmeral Data Fabric (MapR) Hadoop Hive Database SQL DDL Script	Import
IBM Cognos Content Manager Packages	Import
IBM Cognos Framework Manager (FM)	Import/Export
IBM Cognos Framework Manager (FM) for ReportNet	Import/Export
IBM DB2 Cube Views	Import/Export
IBM DB2 for z/OS (mainframe) Database (via JDBC)	Import
IBM DB2 iSeries (AS/400) Database (via JDBC)	Import
IBM DB2 Universal Database (UDB) (via JDBC)	Import
IBM Informix Database (via JDBC)	Import
IBM InfoSphere Data Architect (IDA)	Import/Export

Application	Mode
IBM InfoSphere DataStage	Import/Export
IBM InfoSphere Federation Server (via JDBC)	Import
IBM InfoSphere Warehouse - InfoSphere Data Architect	Import/Export
IBM Lotus Notes (via JDBC-ODBC)	Import
IBM Netezza Database (via JDBC)	Import
IBM PureData System for Analytics - Netezza Database (via JDBC)	Import
IBM Rational Rose 4.0	Import/Export
IBM Rational Rose 6.0(98i) to 6.5(2000)	Import/Export
IBM Rational Rose 7.x (2000e and newer)	Import/Export
IBM Rational Rose XDE Developer (via Rose MDL)	Export
IBM Rational Software Architect (RSA) (via Rose MDL)	Export
IBM Rational Software Architect (RSA) (via UML 2.x XMI)	Import/Export
IBM Rational Software Modeler (RSM) (via Rose MDL)	Export
IBM Rational Software Architect (RSA) (via UML 2.x XMI)	Import
IBM Telelogic Tau (via UML 1.x XMI)	Import/Export
IDERA ER/Studio Business Architect (via DM1)	Import/Export
IDERA ER/Studio Data Architect	Import/Export
IDERA ER/Studio Repository	Import
Informatica Big Data Management (BDM) Developer	Import/Export
Informatica Big Data Management (BDM) Developer Repository	Import

Application	Mode
Informatica Data Engineering Integration (DEI) Developer	Import/Export
Informatica Data Engineering Integration (DEI) Developer Repository	Import
Informatica PowerCenter	Export
Informatica PowerCenter (File)	Import
Informatica PowerCenter (Repository)	Import
IRI CoSORT RowGen Data Definition File	Export
IRI CoSORT SortCL Data Definition File	Export
JSON File	Import
MarkLogic NoSQL Database	Import
Metadata Excel Format	Import/Export
Micro Focus (Merant) AppMaster Builder	Import
Microsoft Azure Analysis Services (SSAS)	Import
Microsoft Azure Blob Storage File System	Import
Microsoft Azure Cosmos DB NoSQL Database	Import
Microsoft Azure Data Lake Storage (ADLS) Gen1 File System	Import
Microsoft Azure Data Lake Storage (ADLS) Gen2 File System	Import
Microsoft Azure Power BI Dataflow (File) - New Beta Bridge	Import
Microsoft Azure SQL Database (via JDBC)	Import
Microsoft Azure Synapse Analytics (SQL Data Warehouse) (via JDBC)	Import
Microsoft Office Visio Database Diagram (via ERX)	Import

Application	Mode
Microsoft Office Visio UML Class Diagram (via UML 1.x XMI)	Import
Microsoft Power BI Desktop Report (File)	Import
Microsoft SQL Server Analysis Services (SSAS) 7.0 to 2000 (Repository)	Import
Microsoft SQL Server Analysis Services (SSAS) (Repository)	Export
Microsoft SQL Server Analysis Services (SSAS) (via DSV File)	Import/Export
Microsoft SQL Server Data Source View (DSV)	Import/Export
Microsoft SQL Server Integration Services (SSIS) (via DSV File)	Import/Export
Microsoft SQL Server XML Data Reduced (XDR) Schema	Import
Microsoft SQL Server Analysis Services (SSAS) (via DSV File)	Import
Microsoft SQL Server Analysis Services (SSAS) 2005 to 2017 (Repository)	Import
Microsoft SQL Server Analysis Services (SSAS) 2005 to 2019 (XMLA File)	Import
Microsoft SQL Server Analysis Services (SSAS) 2016 to 2019 (Tabular Model File)	Import
Microsoft SQL Server Database (via JDBC)	Import
Microsoft SQL Server Database SQL DDL Script	Import
Microsoft Visual Studio / Modeler 2.0 (via MDL)	Import/Export
MicroStrategy 7.0 to 11.x/2021	Import/Export
MicroStrategy MDX Adapter (via ODBO) - Unsupported EOL Bridge	Import
MongoDB NoSQL Database	Import

Application	Mode
NoMagic MagicDraw (via UML 2.x XMI)	Import/Export
OMG CWM 1.x XMI 1.x	Import/Export
OMG CWM Pre-1.0 XMI 1.1 - Unsupported EOL Bridge	Import/Export
OMG UML 1.x XMI 1.x	Import/Export
OMG UML 2.x XMI 2.x	Import/Export
OpenAPI Specification (OAS)	Import
OpenStack Swift Object Store File System - New Beta Bridge	Import
Oracle Business Intelligence (OBI) Publisher Data Model (XDMZ file) - New Beta Bridge	Import
Oracle Business Intelligence (OBI) Publisher Report (XDOZ file) - New Beta Bridge	Import
Oracle Business Intelligence (OBI) Repository (RPD file)	Import
Oracle Database (via JDBC)	Import
Oracle Database SQL DDL Script	Import
Oracle Data Integrator (ODI)	Import
Oracle Data Modeler (ODM) 4.x	Import
Oracle Data Modeler (ODM) 4.x (via erwin 4 XML)	Export
Oracle Designer	Import/Export
Oracle Hyperion Essbase Integration Services (EIS)	Import/Export
Oracle MySQL Database (via JDBC)	Import
Oracle Warehouse Builder (OWB)	Import/Export
PostgreSQL Database (via JDBC)	Import

Application	Mode
Presto Database (via JDBC)	Import
QlikSense Cloud	Import
QlikSense File	Import
QlikView File	Import
SAS OLAP Server (via ODBO) - Unsupported EOL Bridge	Import
SAS Library	Import
SAS Data Integration	Import
SAP BusinessObjects Designer (File)	Import/Export
SAP BusinessObjects Designer (Repository)	Import
SAP BusinessObjects Information Design Tool (IDT) (Repository)	Import
SAP BusinessObjects Information Design Tool (IDT) (File)	Import/Export
SAP HANA Database (via JDBC)	Import
Salesforce (SFDC) Database	Import
SAP BusinessObjects Crystal Reports XI (File)	Import
SAP BusinessObjects Crystal Reports XI (Repository)	Import
Select SE	Import
Snowflake Database (via JDBC)	Import
Sparx Enterprise Architect (EA) (via UML 2.x XMI)	Import/Export
SAP Data Services (BODI/BODS)	Import
SAP PowerDesigner CDM 6.x	Import
SAP PowerDesigner CDM 7.5 to 16.x	Import

Application	Mode
SAP PowerDesigner CDM 8.0 to 16.x	Import/Export
SAP PowerDesigner OOM 9.x to 15.x (via UML 2.x XMI)	Import
SAP PowerDesigner OOM 9.x to 15.x	Export
SAP PowerDesigner PDM 6.1.x	Import
SAP PowerDesigner PDM 7.5 to 16.x	Import
SAP PowerDesigner PDM 8.x to 16.x	Import/Export
SAP PowerDesigner LDM 15.x to 16.x	Import
SAP PowerDesigner XSM 10.x to 15.x	Export
SAP Sybase ASE Database (via JDBC)	Import
Tableau Data Source and Workbook (File)	Import
Tableau Data Source (File)	Export
Tableau Server Extract (Hyper) Data Store	Import
Tableau Server (Repository)	Import
Talend Data Integration	Import/Export
Talend Data Preparation	Import
Talend Master Data Management (MDM)	Import/Export
Teradata Database (via JDBC)	Import
Teradata Database SQL DDL Script	Import
Teradata Vantage Database (via JDBC)	Import
ThoughtSpot	Import
TIBCO Spotfire Analyst (DXP Format) (File)	Import

Application	Mode
TIBCO Spotfire Binary Data (SBDF Format) (File)	Import
TIBCO Spotfire Server (Repository)	Import
TIBCO Spotfire Text Data (STDF Format) (File)	Import
Tigris ArgoUML (via UML 1.x XMI)	Import/Export
Unicom (Popkin) System Architect (SA) 10.4 to 11.4 (Encyclopedia)	Import/Export
Unicom (Popkin) System Architect (SA) 7.1 to 11.4 (File)	Import/Export
Vertica SQL Data Warehouse (via JDBC)	Import
Visible IE:Advantage 6.1	Import
VMware Tanzu Greenplum Database (via JDBC)	Import
W3C Semantic Web Ontology (OWL/RDF)	Import
W3C Web Service Definition Language (WSDL) - Beta Bridge	Import
W3C XML	Import
W3C XML Document Type Definition (DTD) 1.0	Import
W3C XML Schema Definition (XSD) 1.0	Import
W3C XML Schema Definition (XSD) 1.0 Reverse Engineering	Import
W3C XML Schema Definition (XSD) 1.0 Forward Engineering	Export

- For information about how to use the metadata integration wizards, see the erwin Data Modeler online help.
- For Oracle Data Integrator 11g, the new Java API based architecture is supported.
- For Oracle Warehouse Builder 11.2, the new API architecture for 11.2 (different from 11.1) is supported.

- IBM Rational System Architect 7.1 to 11.x (File) and IBM Rational System Architect 10.4 to 11.x (Encyclopedia) must be installed on your local computer before you use these import and export bridges.